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Distributed fault-tolerant consensus for one-sided Lipschitz multiagent systems based on error decomposition and jointly observable condition

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ABSTRACT

This article studies the performance guaranteed tracking control problem for one-sided Lipschitz (OSL) multiagent systems (MASs) with unknown actuator faults under jointly observable condition (JOC). Unlike the recent results where the state or output information of the leader should be known to its neighbors, only partial output information of the leader can be obtained by its neighbors under JOC in this article. One challenge focuses on the distributed observer design with OSL dynamics under JOC. The studied problem is proceeded in two steps. Firstly, to deal with the difficulty caused by OSL dynamics, an error decomposition technique is used to design the distributed observer. Secondly, based on this observer, a distributed adaptive fault-tolerant protocol with guaranteed performance is designed to realize tracking task. Note that unlike the existing ideas, actuator faults of an agent will not have effect on the dynamics of its neighbors over the communication graph in this article. Simulation results illustrate the developed algorithm.

1. Introduction

In the past decades, distributed control of multiagent systems (MASs) has attracted much attention due to lots of practical applications in sensor networks, oscillator networks, unmanned aerial vehicles, and so on [1–5]. A critical class of cooperative control problem is the leader-following consensus or distributed tracking control. The aim is to drive all followers to track the trajectories of the leader in a distributed manner [6].

Recently, many attractive works have been reported on distributed control of MASs [7–10]. Note that the aforementioned contributions [5,7–10] assume that output or state information of the leader is exactly known to its neighbors. However, it is worth noting that in real-world applications, owing to various factors such as physical or geographical restrictions, it is usually difficult for neighbors of the leader to directly access the full state or output information of the leader. In fact, a single agent might be able to access only partial output of the leader [11]. Consequently, the above distributed state estimation methods in [7–10] do not directly apply anymore. Therefore, it is desirable to relax the limitation in above results, and the interest in investigating the problem is growing

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as in [12,13]. In [12], the JOC is introduced, in which neighbors of the leader can only obtain the partial output information of the leader. This condition is a more relaxed assumption for the distributed state observer design. By virtue of the observability decomposition, a distributed state observer of linear system is proposed with JOC [12]. More recently, this attractive work has been extended in [13–15]. In [13], by constructing an event-triggered distributed observer, the problem of output tracking control is handled. In [15], the distributed unknown input observers are proposed to reconstruct the state of the target systems with unknown inputs. It should be noted that in [12–15], the distributed observer design relies on the observability decomposition, these methods can only deal with the case that the target system is linear. In this article, we consider a more general target system containing OSL nonlinear dynamics. To deal with the difficulty caused by OSL nonlinear dynamics, observability decomposition and an error decomposition technique are used to design the distributed OSL nonlinear observers in this article.

In general, owing to the influence of environment and system characteristics, there exist certain nonlinearities in physical systems from a practical perspective, thus it is meaningful to study the nonlinear MASs. Most nonlinearities satisfy the Lipschitz condition, at least locally, which causes that the Lipschitz nonlinear MASs have been widely studied in the existing results [16–19]. However, the restriction of all these methods lies in the fact that if the Lipschitz constant is large, the provided results are unable to handle this case [20,21]. To relax the conservatism of the Lipschitz condition and deal with a larger class of nonlinear systems, a more realistic and general condition is considered, which is called the OSL condition [20]. It has been verified that if a nonlinear function satisfies the Lipschitz condition, then the OSL constant is smaller or equal to Lipschitz constant [20]. In recent years, distributed control for Lipschitz systems has been extended to more general cases in the presence of OSL nonlinearities in dynamics of systems [22–29]. In [22], for nonlinear OSL stochastic systems, the problem of observer design is investigated, which indicates that the exponential convergence of the observer's error could be obtained. For OSL MASs, distributed consensus control methods are constructed under directed and undirected graphs, respectively [23,24]. To reduce communication burden, dynamics event-triggered control schemes are designed to realize leaderless consensus [25–27]. In [29], the leader-following impulsive consensus problem of OSL MASs is addressed under switching networks. It is worth noting that in the aforementioned works for OSL MASs, neighbors of the leader rely on all output or state information of the leader. It is challenging to design the distributed observer under JOC in the presence of OSL dynamics.

On the other hand, the occurrence of system failures is inevitable due to long-term utilization, wear, unpredictable change of external environment and so on, which might cause system performance degradation or even system instability [30–33]. Thus, the fault-tolerant (FT) control of MASs is a meaningful problem. Recently, lots of fruitful results on FT control issue of MASs have been reported [34–37]. By designing adaptive gains, the distributed FT control scheme for homogeneous linear MASs is proposed [35]. In [36], for heterogeneous nonlinear MASs, the distributed adaptive FT fuzzy tracking control scheme is proposed to compensate the impact of actuator faults. It should be noted that although the FT control strategy is designed, the actuator faults of an agent will have effect on the performance of its neighbors, which is caused by the exchange of state or output information among neighbor agents over the communication network [35,36]. To deal with this problem, a hierarchical control scheme is proposed in [37], the first layer is to design a virtual system by constructing the distributed observer, the second layer is composed of an actual system and a FT controller to track the first layer virtual system. By virtue of this method, the distributed FT control issue of heterogeneous linear MASs with multiple leaders is handled. However, it is worth emphasizing that in the above FT control works, OSL dynamics of the leader are not considered, and the full state or output information of the leader should be known to its neighbors.

Motivated by the works mentioned above, under directed communication graph and JOC, the distributed adaptive FT tracking control issue is addressed for the OSL MASs subject to unknown actuator faults. The crucial contributions of this article are presented as follows.

1. A distributed observer-based FT control scheme is designed, which can avoid actuator faults propagation among neighbors. The studied problem is decoupled into two parts, namely, the problem of developing the distributed OSL observer under JOC, and the FT performance guaranteed tracking control problem of all OSL followers.
2. An error decomposition technique is introduced to design the distributed observer under JOC when the leader's model contains OSL nonlinearity. This is a step ahead from the existing results under JOC [12–14], which are only applicable to linear leader's model.
3. Those previous FT control results [37–39] with observability conditions and the linear leader's model can be regarded as special cases of our results, which indicates that our results can be applied more widely. Besides, the generalized restricted potential function is used to improve the transient performance under actuator faults.

The organization of this article is organized as follows. Section 2 presents preliminaries and problem formulation. Section 3 is focused on constructing distributed observer and distributed observer-based FT controller. A simulation example is shown in Section 4. Finally, Section 5 presents the conclusion.

2. System description and preliminary knowledge

2.1. Notations and graph theory

For a matrix $\bar{Q}\in R^{p\times q}$, $\ker(\bar{Q}) = x\in R^q|\bar{Q}x = 0$ and $\mathrm{im}(\bar{Q}) = \{z\in R^p|z = \bar{Q}x, \text{for some } x\in R^q\}$ represent the kernel and image of $\bar{Q}$, respectively. For a subspace $\bar{\mathcal{P}}\subset R^n$, the orthogonal complement of $\bar{\mathcal{P}}$ is denoted as $\bar{\mathcal{P}}^{\perp}$. Let $1_N = [1, ..., 1]^T \in R^N$, 0 be the zero matrix with appropriate dimensions. For a set of matrices $\left\{\iota_i \in R^{g_i\times n} \mid i = 1, 2, \dots, N\right\}$ and a set $\mathcal{N} = \{1, ..., N\}$, let

$\mathrm{col}(\iota_i)_{i\in\mathcal{N}} = [\iota_1^T, \iota_2^T, ..., \iota_N^T]^T$ and $\mathrm{row}(\iota_i)_{i\in\mathcal{N}} = [\iota_1, \iota_2, ..., \iota_N]$. For a set of matrices $\{\bar{\iota}_i \in R^{n\times n} \mid i = 1, 2, \dots, N\}$ and a set $\mathcal{N} = \{1, ..., N\}$, let $\mathrm{diag}(\bar{\iota}_1, \bar{\iota}_2, ..., \bar{\iota}_N)$ denote the block diagonal matrices of $\bar{\iota}_i$. For a matrix $A \in R^{n\times n}$, let $\lambda(A)$ denote the eigenvalues of A, $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$ represent the biggest and smallest eigenvalues of A separately. $I_n \in R^{n\times n}$ denotes identity matrix.

The communication topology among N agents is denoted by a graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N} = \{1, ..., N\}$ denotes the set of nodes, and $\mathcal{E} = \mathcal{N} \times \mathcal{N}$ represents the set of edges. An edge (i, j) indicates that agent i can send the information to agent j. The adjacency matrix of the graph $\mathcal{G}$ is denoted by $A = [a_{ij}] \in R^{N\times N}$, which is defined by $a_{ij} = 1$ if $(j, i) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. The Laplacian matrix $\mathcal{L} = [l_{ij}] \in R^{N\times N}$ of graph $\mathcal{G}$ is defined as $l_{ij} = -a_{ij}$ and $l_{ii} = \sum_{j=1, j\neq i}^{n} a_{ij}$. For $\forall i, j \in \mathcal{N}$, if there exists a directed path from node i to node j, then graph $\mathcal{G}$ is strongly directed.

Assumption 1. The communication graph $\mathcal{G}$ is strongly connected directed graph.

Lemma 1. *[12] Let $\mathcal{G}$ be a strongly connected directed graph. Then, it results that 1) there exists a unique vector $s = [s_1, ..., s_N]^T$ with $s_i > 0$ such that $s^T 1_N = N$ and $s^T \mathcal{L} = 0$. 2) Let $S = \mathrm{diag}\{s_1, s_2, ..., s_N\}$, then $\hat{\mathcal{L}} = S\mathcal{L} + \mathcal{L}^T S$ is symmetric matrix and can be regarded as the Laplacian matrix associated with an undirected connected graph $\hat{\mathcal{G}}$ obtained from $\mathcal{G}$ by ignoring the directions of the edges.*

2.2. System dynamics

Consider a group of agents with N followers and a leader. The dynamics of ith follower can be expressed by

$$\dot{x}_i = \mathcal{A}x_i + \mathcal{B}u_i + \phi(x_i) \tag{1}$$

where $x_i \in R^n$, $u_i \in R^m$ and $\phi(x_i) \in R^n$ are the state, control input and nonlinear function separately. Both $\mathcal{A} \in R^{n\times n}$ and $\mathcal{B} \in R^{n\times m}$ are constant matrices.

In addition, the dynamics of the leader are given by

$$\dot{x}_0 = \mathcal{A}x_0 + \phi(x_0) \tag{2}$$

where let $y_0 = \bar{\mathcal{J}}x_0$ be the output of the leader, $\bar{\mathcal{J}} \in R^{q\times n}$ is constant matrix. In this article, neighbors of the leader can only observe partial output information $\tilde{y}_i = \mathcal{J}_i x_0$ of the system (2), where $\mathcal{J}_i \in R^{\bar{p}_i \times n}$ is constant matrix.

According to [38,37], actuator faults are considered as

$$u_i^F = \pi_i u_i(t) + \rho_i(t) \tag{3}$$

where $\pi_i = (I_m - \chi(\delta_i))$ is the effectiveness matrix of actuator, $\delta_i = [\delta_{i1}, ..., \delta_{im}]^T$, $\chi(\delta_i) = \mathrm{diag}(\delta_{i1}, ..., \delta_{im})$. Unknown constant δ_{ij} is the fault factor of loss effectiveness, which satisfies $\underline{\delta}_{ij} < \delta_{ij} \le \bar{\delta}_{ij} < 1$, bounded constants $\underline{\delta}_{ij}$ and $\bar{\delta}_{ij}$ are the unknown bounds of δ_{ij}. $\rho_i(t) = [\rho_{i1}, ..., \rho_{im}]^T$ is the unknown bias fault, which satisfies $\|\rho_i(t)\| \le \vartheta_i$, ϑ_i is the unknown bounded constant. Then, the dynamics of follower i with actuator faults are given as

$$\dot{x}_i = \mathcal{A}x_i + \mathcal{B}(\pi_i u_i(t) + \rho_i(t)) + \phi(x_i) \tag{4}$$

Definition 1. If there exists $\gamma_1 \in R$, e, $\bar{e} \in R^n$ such that the function $\phi(\cdot)$ satisfies

$$\langle \phi(e) - \phi(\bar{e}), e - \bar{e} \rangle \le \gamma_1 \|e - \bar{e}\|^2 \tag{5}$$

It is said that $\phi(\cdot)$ is one-sided Lipschitz (OSL) function.

Definition 2. For the function $\phi(\cdot)$, if there exist γ_2, $\gamma_3 \in R$, e, $\bar{e} \in R^n$ such that

$$(\phi(e) - \phi(\bar{e}))^T(\phi(e) - \phi(\bar{e})) \le \gamma_2 \|e - \bar{e}\|^2 + \gamma_3 \langle \phi(e) - \phi(\bar{e}), e - \bar{e} \rangle \tag{6}$$

It is said that the function $\phi(\cdot)$ satisfies the quadratic inner-boundedness (QIB) condition.

Assumption 2. The nonlinear function $\phi(\cdot)$ in (1) and (2) satisfies Definitions 1 and 2.

Assumption 3 *(Jointly observable condition).* The pair $(\mathcal{A}, \mathrm{col}(\mathcal{J}_i)_{i\in\mathcal{N}})$ is observable.

Remark 1. OSL and QIB conditions in Assumption 2 are common in OSL systems results, which generalizes the classical Lipschitz condition to a more general class of nonlinear systems [7], [8]. It should be noted that the Lipschitz constant must be zero or positive. However, the parameters γ_1, γ_2 and γ_3 can be positive, zero or negative which can expand the solution's feasibility region. If a function is Lipschitz, then it also satisfies OSL and QIB conditions, but the converse is not true.

Remark 2. Assumption 3 is the jointly observable condition (JOC), which is used in the design of the distributed observer [12–15,40]. Note that the above results are limited to the case that the dynamics of the leader are linear. This is the reason that motivate us to investigate more general OSL nonlinear MASs, which is a step ahead from the above results. Besides, the pair $(\mathcal{A}, \mathcal{J}_i)$ is not necessarily observable, which indicates that the local measurement of each follower might not be able to estimate the state of the leader.

Since the $(\mathcal{A}, \mathcal{J}_i)$ might be unobservable, the observability decomposition for $(\mathcal{A}, \mathcal{J}_i)$ should be done. Now, based on observability decomposition, $\mathcal{A}$ and $\mathcal{J}_i$ are transformed into the following form

$$\mathcal{M}_i^T \mathcal{A} \mathcal{M}_i = \begin{bmatrix} \mathcal{A}_{io} & 0 \\ \mathcal{A}_{ir} & \mathcal{A}_{iu} \end{bmatrix}, \quad \mathcal{J}_i \mathcal{M}_i \quad = \begin{bmatrix} \mathcal{J}_{io} & 0 \end{bmatrix} \tag{7}$$

where $\mathcal{A}_{io} \in^{v_i \times v_i}$, $\mathcal{A}_{ir} \in^{(n-v_i) \times v_i}$, $\mathcal{A}_{iu} \in^{(n-v_i)\times(n-v_i)}$, $\mathcal{J}_{io} \in^{\bar{p}_i \times v_i}$, and $n - v_i$ is the dimension of the unobservable subspace of the pair $(\mathcal{A}, \mathcal{J}_i)$, the pair $(\mathcal{A}_{io}, \mathcal{J}_{io})$ is observable. Moreover, $\mathcal{M}_i = [\mathcal{M}_{io}, \mathcal{M}_{iu}]$ is orthogonal matrix. The unobservable subspace is given by $\mathrm{im}(\mathcal{M}_{iu}) = \ker(\mathcal{X}_i)$, where $\mathcal{X}_i = \mathrm{col}(\mathcal{J}_i \mathcal{A}^{o-1})_{o \in \bar{\mathcal{N}}}$, $\bar{\mathcal{N}} = \{1, \ldots, n\}$. Note that $\mathrm{im}(\mathcal{M}_{io}) = \ker(\mathcal{X}_i)^{\perp}$.

2.3. Generalized restricted potential function

To guarantee time-varying performance bounds in the presence of actuator faults, the generalized restricted potential function (GRPF) [41,42] is introduced.

For given vector $\xi \in R^q$ and matrix $\mathcal{W} \in R_+^{q\times q}$, let $\|\xi\|_{\mathcal{W}} = \sqrt{\xi^T \mathcal{W} \xi}$. $\psi(\|\xi\|_{\mathcal{W}})$, $\psi : R^q \to R$ is a generalized restricted potential function on the set $\mathcal{I}_\theta \triangleq \left\{\|\xi\|_{\mathcal{W}} : \|\xi\|_{\mathcal{W}} \in [0, \theta(t))\right\}, \theta(t) > 0$, if the following statements hold

1) If $\|\xi\|_{\mathcal{W}} = 0$, then $\psi(\|\xi\|_{\mathcal{W}}) = 0$.
2) If $\|\xi\|_{\mathcal{W}} \in \mathcal{I}_\theta$ and $\|\xi\|_{\mathcal{W}} \neq 0$, then $\psi(\|\xi\|_{\mathcal{W}}) > 0$.
3) If $\|\xi\|_{\mathcal{W}} \to \theta(t)$, then $\psi(\|\xi\|_{\mathcal{W}}) \to \infty$.
4) $\psi(\|\xi\|_{\mathcal{W}})$ is continuously differentiable on $\mathcal{I}_\theta$.
5) If $\|\xi\|_{\mathcal{W}} \in \mathcal{I}_\theta$, then $\psi_d(\|\xi\|_{\mathcal{W}}) > 0$, where

$$\psi_d(\|\xi\|_{\mathcal{W}}) = \frac{\mathrm{d}\psi\left(\|\xi\|_{\mathcal{W}}\right)}{\mathrm{d}\|\xi\|_{\mathcal{W}}^2}. \tag{8}$$

6) If $\psi(\|\xi\|_{\mathcal{W}}) \in \mathcal{I}_\theta$, then $2\psi_d(\|\xi\|_{\mathcal{W}})\|\xi\|_{\mathcal{W}}^2 - \psi(\|\xi\|_{\mathcal{W}}) > 0$.

In this article, $\psi(\|\xi\|_{\mathcal{W}})$ is chosen as

$$\psi(\|\xi\|_{\mathcal{W}}) = \frac{\|\xi\|_{\mathcal{W}}^2}{\theta^2(t) - \|\xi\|_{\mathcal{W}}^2}, \quad \|\xi\|_{\mathcal{W}} \in \mathcal{I}_\theta \tag{9}$$

From (8) and (9), it can be concluded that $\psi_d(\|\xi\|_{\mathcal{W}}) = \frac{\theta^2(t)}{\left(\theta^2(t) - \|\xi\|_{\mathcal{W}}^2\right)^2} > 0$ and $2\psi_d(\|\xi\|_{\mathcal{W}})\|\xi\|_{\mathcal{W}}^2 - \psi(\|\xi\|_{\mathcal{W}}) = \frac{\theta^2(t)\|\xi\|_{\mathcal{W}}^2 + \|\xi\|_{\mathcal{W}}^4}{\left(\theta^2(t) - \|\xi\|_{\mathcal{W}}^2\right)^2} > 0, \quad \|\xi\|_{\mathcal{W}} \in \mathcal{I}_\theta$. Thus, equation (9) satisfies all statements 1)-6).

2.4. Control objective

To propose a distributed fault-tolerant tracking control scheme such that $\lim_{t\to\infty}\left(x_i - x_0\right) = 0, i \in \mathcal{N}$.

3. Distributed observer-based fault-tolerant performance guaranteed control design

3.1. Design of distributed observer

In this subsection, based on the partial information $\tilde{y}_i$ and relative estimated state information among neighbors, a distributed observer is constructed to estimate the full state of leader under a directed graph. For ith follower, let ζ_i be the estimate state of the leader. Here, the distributed observer is as follows:

$$\dot{\zeta}_i = \mathcal{A}\zeta_i + \phi(\zeta_i) + \mathcal{Y}_i\left(\mathcal{J}_i\zeta_i - \tilde{y}_i\right) + \sigma_0 s_i \mathcal{F}_i^{-1} \sum_{j=1}^N a_{ij}\left(\zeta_j - \zeta_i\right) \tag{10}$$

where $\mathcal{Y}_i$ and $\mathcal{F}_i^{-1}$ are matrices to be designed. σ_0 is a positive constant. In addition, s_i is given in Lemma 1.

Define the estimation error $\varpi_i = \zeta_i - x_0$, the dynamics of estimation error satisfy

$$\dot{\varpi}_i = \left(\mathcal{A} + \mathcal{Y}_i\mathcal{J}_i\right)\varpi_i + \bar{\phi}_i + \sigma_0 s_i \mathcal{F}_i^{-1} \sum_{j=1}^N a_{ij}\left(\varpi_j - \varpi_i\right) \tag{11}$$

where $\bar{\phi}_i = \phi(\zeta_i) - \phi(x_0)$.

Lemma 2. *Consider the distributed observer (10) under Assumptions 1-3. For given positive constants κ_1, κ_2 and σ_0, if there exist matrices $\mathcal{F}_{io} = \mathcal{F}_{io}^T(\mathcal{F}_{io} > 0)$, $\mathcal{F}_{iu} = \mathcal{F}_{iu}^T(\mathcal{F}_{iu} > 0)$ and $\mathcal{H}_i$ with appropriate dimensions such that the following linear matrix inequality (LMI) holds*

$$\begin{bmatrix} \Pi_0 & -\Gamma_{\mathcal{F}} & \Xi_{\mathcal{F}} \\ -\Gamma_{\mathcal{F}}^T & \Pi_1 & \Xi \\ \Xi_{\mathcal{F}}^T & \Xi^T & \kappa_2 I_{Nn} \end{bmatrix} > 0 \tag{12}$$

where

$$\Pi_0 = -\sum_{i=1}^{N} \Gamma_i - c_1 I_n, \quad \Gamma_i = \left(\mathcal{A} + \mathcal{Y}_i \mathcal{J}_i\right)^T \mathcal{F}_i + \mathcal{F}_i \left(\mathcal{A} + \mathcal{Y}_i \mathcal{J}_i\right)$$

$$\mathcal{Y}_i = \mathcal{M}_i \begin{bmatrix} \mathcal{F}_{io}^{-1} \mathcal{J}_{io}^T \mathcal{H}_i^T \\ 0_{(n-v_i)\times \bar{p}_i} \end{bmatrix}, \quad \mathcal{F}_i = \mathcal{M}_i \begin{bmatrix} \mathcal{F}_{io} & 0_{v_i \times (n-v_i)} \\ 0_{(n-v_i)\times v_i} & \mathcal{F}_{iu} \end{bmatrix} \mathcal{M}_i^T, \quad \Pi_1 = \sigma_0 \sigma_1 I_{Nn} - \Gamma - c_2 I_{Nn}$$

$$\Gamma_{\mathcal{F}} = \mathrm{row}(\Gamma_i)_{i\in\mathcal{N}}, \quad \Xi_{\mathcal{F}} = \mathrm{row}(\Xi_i)_{i\in\mathcal{N}}, \quad \Gamma = \mathrm{diag}(\Gamma_i)_{i\in\mathcal{N}}, \quad \Xi = \mathrm{diag}(\Xi_i)_{i\in\mathcal{N}}$$

with $\Xi_i = -(\mathcal{F}_i + c_3 I_n)$*,* $c_1 = (\kappa_1\gamma_1 + \kappa_2\gamma_2)N$*,* $c_2 = (\kappa_1\gamma_1 + \kappa_2\gamma_2)$*,* $c_3 = 0.5(-\kappa_1 + \kappa_2\gamma_3)$*.* σ_1 *is the smallest nonzero eigenvalue of* $\hat{\mathcal{L}}$*. Hence, the estimation error* $\varpi_i = \zeta_i - x_0$ *exponentially converges to zero.*

Proof. Construct Lyapunov function $\mathcal{U}_o$ as

$$\mathcal{U}_o = \sum_{i=1}^{N} \varpi_i^T \mathcal{F}_i \varpi_i \tag{13}$$

From (11) and (13), the time derivation of $\mathcal{U}_o$ yields

$$\begin{aligned} \dot{\mathcal{U}}_o &= \sum_{i=1}^{N} \varpi_i^T \Gamma_i \varpi_i + 2\sigma_0 \sum_{i=1}^{N}\sum_{j=1}^{N} a_{ij} s_i \left(\varpi_j - \varpi_i\right)^T \varpi_i \\ &\quad - 2\sum_{i=1}^{N} \varpi_i^T \mathcal{F}_i \bar{\phi}_i \\ &= \varpi^T \Gamma \varpi - \sigma_0 \varpi^T (\hat{\mathcal{L}} \otimes I_n)\varpi + 2\varpi^T \bar{\mathcal{F}} \bar{\phi} \end{aligned} \tag{14}$$

where $\hat{\mathcal{L}} = S\mathcal{L} + \mathcal{L}^T S$, $\bar{\mathcal{F}} = \mathrm{diag}(\mathcal{F}_i)_{i\in\mathcal{N}}$. $\bar{\phi} = \mathrm{col}(\bar{\phi}_i)_{i\in\mathcal{N}}$, $\varpi = \mathrm{col}(\varpi_i)_{i\in\mathcal{N}}$.

Remark 3. It is worth noting that $(\mathcal{A}, \mathcal{J}_i)$ might be unobservable, thus $\varpi^T \Gamma \varpi$ in (14) might not be negative definite. Owing to the term $\bar{\phi}$ with $\bar{\phi}_i = \phi(\zeta_i) - \phi(x_0)$ in (14), these methods in [12–14] are no longer applicable. In light of Assumption 1 and Lemma 1, the Laplacian matrix $\hat{\mathcal{L}}$ has a zero eigenvalue and positive eigenvalues, thus the term $-\sigma_0 \varpi^T (\hat{\mathcal{L}} \otimes I_n)\varpi$ is negative semidefinite. To guarantee that the right-hand side of (14) is negative definite, the space of error term ϖ is decomposed into two complementary subspaces, namely agreement subspace and disagreement subspace. Such decomposition utilizes the property of the Laplacian matrix of $\hat{\mathcal{L}}$, namely $\hat{\mathcal{L}} \times 1_N = 0$. Base on the above decomposition, a negative definite term is induced from the negative semidefinite term $-\sigma_0 \varpi^T (\hat{\mathcal{L}} \otimes I_n)\varpi$.

To deal with (14), the estimate error $\varpi \in R^{Nn}$ is decomposed as

$$\varpi = \varpi_c + \varpi_r \tag{15}$$

where the error space R^{Nn} is denoted as $\Omega_0 \oplus \Omega_0^{\perp}$. Ω_0 is the agreement set, which is defined as $\Omega_0 = \left\{\varpi \in R^{Nn} : \varpi_i = \varpi_j, \forall i, j \in \mathcal{N}\right\}$. $\varpi_c \in \Omega_0$ with the form $\varpi_c = 1_N \otimes \omega$, $\omega \in R^n$. Besides, $\varpi_r \in \Omega_0^{\perp}$ and $\varpi_c^T \varpi_r = 0$.

According to (15) and (16), it implies that

$$\begin{aligned} \dot{\mathcal{U}}_o &= (\varpi_c + \varpi_r)^T \Gamma (\varpi_c + \varpi_r) - \sigma_0 (\varpi_c + \varpi_r)^T (\hat{\mathcal{L}} \otimes I_n)(\varpi_c + \varpi_r) \\ &\quad + 2(\varpi_c + \varpi_r)^T \bar{\mathcal{F}} \bar{\phi} \\ &= \varpi_c^T \Gamma \varpi_c + \varpi_r^T \Gamma \varpi_r + 2\varpi_c^T \Gamma \varpi_r - \sigma_0 \varpi_r^T (\hat{\mathcal{L}} \otimes I_n)\varpi_r \\ &\quad + 2\varpi_c^T \bar{\mathcal{F}} \bar{\phi} + 2\varpi_r^T \bar{\mathcal{F}} \bar{\phi} \end{aligned} \tag{16}$$

Based on the definitions of ϖ_r and ϖ_c, it indicates from (16) that

$$\begin{cases} \varpi_c^T \Gamma \varpi_c = (1_N \otimes \omega)^T \Gamma (1_N \otimes \omega) = \omega^T \left(\sum_{i=1}^{N} \Gamma_i\right) \omega \\ \varpi_r^T \Gamma \varpi_c = \varpi_r^T \Gamma^T (1_N \otimes \omega) = \varpi_r^T \Gamma_{\mathcal{F}}^T \omega = \omega^T \Gamma_{\mathcal{F}} \varpi_r \\ \varpi_c^T \bar{\mathcal{F}} \bar{\phi} = (1_N \otimes \omega)^T \bar{\mathcal{F}} \bar{\phi} = \bar{\phi} \tilde{\mathcal{F}}^T \omega = \omega^T \tilde{\mathcal{F}} \bar{\phi} \end{cases} \tag{17}$$

where $\tilde{\mathcal{F}} = \mathrm{row}(\mathcal{F}_i)_{i\in\mathcal{N}}$.

Moreover, based on $\varpi_r^T \varpi_c = 0$, it results that $\varpi_r^T (1_N \otimes \omega) = 0$, where $\omega \in R^n$. Under Assumption 1 and Lemma 1, it results from [43] that

$$-\varpi_r^T\left(\hat{\mathcal{L}}\otimes I_n\right)\varpi_r\leq-\sigma_1\varpi_r^T\varpi_r \tag{18}$$

Substitute (17) and (18) into (16) yields

$$\begin{aligned}\dot{\mathcal{V}}_o=&\,\omega^T\left(\sum_{i=1}^N\Gamma_i\right)\omega+\varpi_r^T\Gamma\varpi_r+2\omega^T\Gamma_P\varpi_r\\&-\sigma_0\sigma_1\varpi_r^T\varpi_r+2\omega^T\tilde{\mathcal{F}}\bar{\phi}+2\varpi_r^T\bar{\mathcal{F}}\bar{\phi}\end{aligned} \tag{19}$$

According to Definitions 1 and 2 and Assumption 2, it indicates that

$$\kappa_1\gamma_1\sum_{i=1}^N\varpi_i^T\varpi_i-\kappa_1\sum_{i=1}^N\varpi_i^T\bar{\phi}_i\geq 0 \tag{20a}$$

$$-\kappa_2\sum_{i=1}^N\bar{\phi}_i^T\bar{\phi}_i+\kappa_2\gamma_2\sum_{i=1}^N\varpi_i^T\varpi_i+\kappa_2\gamma_3\sum_{i=1}^N\varpi_i^T\bar{\phi}_i\geq 0 \tag{20b}$$

where design parameters $\kappa_1(\kappa_1>0)$ and $\kappa_2>0(\kappa_2>0)$ are introduced to increase the freedom of distributed observer design.

From (19), (20a) and (20b), it holds that

$$\begin{aligned}\dot{\mathcal{V}}_o\leq&\,\omega^T\left(\sum_{i=1}^N\Gamma_i\right)\omega+\varpi_r^T\Gamma\varpi_r+2\omega^T\Gamma_{\mathcal{F}}\varpi_r-\sigma_0\sigma_1\varpi_r^T\varpi_r+2\omega^T\tilde{\mathcal{F}}\bar{\phi}+2\varpi_r^T\bar{\mathcal{F}}\bar{\phi}\\&+\kappa_1\gamma_1\sum_{i=1}^N\varpi_i^T\varpi_i-\kappa_1\sum_{i=1}^N\varpi_i^T\bar{\phi}_i-\kappa_2\sum_{i=1}^N\bar{\phi}_i^T\bar{\phi}_i+\kappa_2\gamma_2\sum_{i=1}^N\varpi_i^T\varpi_i\\&+\kappa_2\gamma_3\sum_{i=1}^N\varpi_i^T\bar{\phi}_i\\\leq&\,\omega^T\left(\sum_{i=1}^N\Gamma_i\right)\omega+\varpi_r^T\Gamma\varpi_r+2\omega^T\Gamma_{\mathcal{F}}\varpi_r-\sigma_0\sigma_1\varpi_r^T\varpi_r+2\omega^T\tilde{\mathcal{F}}\bar{\phi}+2\varpi_r^T\bar{\mathcal{F}}\bar{\phi}\\&+\kappa_1\gamma_1\varpi^T\varpi+\kappa_2\gamma_2\varpi^T\varpi+\kappa_2\gamma_3\varpi^T\bar{\phi}-\kappa_2\bar{\phi}^T\bar{\phi}-\kappa_1\varpi^T\bar{\phi}\end{aligned} \tag{21}$$

Since $\varpi=\varpi_c+\varpi_r$, then (21) can be written as

$$\begin{aligned}\dot{\mathcal{V}}_o\leq&\,\omega^T\left(\sum_{i=1}^N\Gamma_i\right)\omega+\varpi_r^T\Gamma\varpi_r+2\omega^T\Gamma_{\mathcal{F}}\varpi_r-\sigma_0\sigma_1\varpi_r^Te_r+2\omega^T\tilde{\mathcal{F}}\bar{\phi}+2\varpi_r^T\bar{\mathcal{F}}\bar{\phi}\\&+\kappa_1\gamma_1(\varpi_c+\varpi_r)^T(\varpi_c+\varpi_r)+\kappa_2\gamma_2(\varpi_c+\varpi_r)^T(\varpi_c+\varpi_r)\\&+\kappa_2\gamma_3(\varpi_c+\varpi_r)^T\bar{\phi}-\kappa_2\bar{\phi}^T\bar{\phi}-\kappa_1(\varpi_c+\varpi_r)^T\bar{\phi}\\\leq&\,\omega^T\left(\sum_{i=1}^N\Gamma_i\right)\omega+\varpi_r^T\Gamma\varpi_r+2\omega^T\Gamma_{\mathcal{F}}\varpi_r-\sigma_0\sigma_1\varpi_r^T\varpi_r+2\omega^T\tilde{\mathcal{F}}\bar{\phi}+2\varpi_r^T\bar{\mathcal{F}}\bar{\phi}\\&+\kappa_1\gamma_1\varpi_c^T\varpi_c+2\kappa_1\gamma_1\varpi_c^T\varpi_r+\kappa_1\gamma_1\varpi_r^T\varpi_r+\kappa_2\gamma_2\varpi_c^T\varpi_c+2\kappa_2\gamma_2\varpi_c^T\varpi_r\\&+\kappa_2\gamma_2\varpi_r^T\varpi_r+\kappa_2\varpi_c^T\bar{\phi}+\kappa_2\varpi_r^T\bar{\phi}-\kappa_2\bar{\phi}^T\bar{\phi}-\kappa_1\varpi_c^T\bar{\phi}-\kappa_1\varpi_r^T\bar{\phi}\\\leq&\,\omega^T\left(\sum_{i=1}^N\Gamma_i\right)\omega+\varpi_r^T\Gamma\varpi_r+2\omega^T\Gamma_{\mathcal{F}}\varpi_r-\sigma_0\sigma_1\varpi_r^T\varpi_r+2\omega^T\tilde{\mathcal{F}}\bar{\phi}+2\varpi_r^T\bar{\mathcal{F}}\bar{\phi}\\&+\kappa_1\gamma_1N\omega^T\omega+\kappa_1\gamma_1\varpi_r^T\varpi_r+\kappa_2\gamma_2N\omega^T\omega+\kappa_2\gamma_2\varpi_r^T\varpi_r\\&+\kappa_2\gamma_3\varpi_c^T\bar{\phi}+\kappa_2\gamma_3\varpi_r^T\bar{\phi}-\kappa_2\bar{\phi}^T\bar{\phi}-\kappa_1\varpi_c^T\bar{\phi}-\kappa_1\varpi_r^T\bar{\phi}\\\leq&\,\omega^T\left(\sum_{i=1}^N\Gamma_i\right)\omega+\varpi_r^T\Gamma\varpi_r+2\omega^T\Gamma_{\mathcal{F}}\varpi_r-\sigma_0\sigma_1\varpi_r^T\varpi_r+2\omega^T\tilde{\mathcal{F}}\bar{\phi}+2\varpi_r^T\bar{\mathcal{F}}\bar{\phi}\\&+c_1\omega^T\omega+c_2\varpi_r^T\varpi_r+2c_3\varpi_c^T\bar{\phi}+2c_3\varpi_r^T\bar{\phi}-\kappa_2\bar{\phi}^T\bar{\phi}\end{aligned} \tag{22}$$

where $c_1=(\kappa_1\gamma_1+\kappa_2\gamma_2)N$, $c_2=(\kappa_1\gamma_1+\kappa_2\gamma_2)$, $c_3=0.5(-\kappa_1+\kappa_2\gamma_3)$, $\varpi_c=1_N\otimes\omega$, $\varpi_c^T\varpi_r=0$, and

$$\begin{cases}\varpi_c^T\varpi_c=(1_N\otimes\omega)^TI_{Nn}(1_N\otimes\omega)=N\omega^T\omega\\2c_3\varpi_c^T\bar{\phi}=(1_N\otimes\omega)^TI_{Nn}\bar{\phi}=2c_3\bar{\phi}\Delta^T\omega=2c_3\omega^T\Delta\bar{\phi}\end{cases} \tag{23}$$

where $\Delta=[I_n,I_n,\ldots,I_n]\in R^{n\times nN}$.

In light of (22) and (23), it implies that

$$\begin{aligned}\dot{\mathcal{V}}_o &\leq \omega^T\left(\sum_{i=1}^{N}\Gamma_i\right)\omega + \varpi_r^T\Gamma\varpi_r + 2\omega^T\Gamma_{\mathcal{F}}\varpi_r - \sigma_0\sigma_1\varpi_r^T\varpi_r + 2\omega^T\tilde{\mathcal{F}}\bar{\phi} + 2\varpi_r^T\bar{\mathcal{F}}\bar{\phi}\\ &\quad + c_1\omega^T\omega + c_2\varpi_r^T\varpi_r + 2c_3\varpi_c^T\Delta\bar{\phi} + 2c_3\varpi_r^T\bar{\phi} - \kappa_2\bar{\phi}^T\bar{\phi}\\ &\leq \omega^T\left(\sum_{i=1}^{N}\Gamma_i\right)\omega + \varpi_r^T\Gamma\varpi_r + 2\omega^T\Gamma_{\mathcal{F}}\varpi_r - \sigma_0\sigma_1\varpi_r^T\varpi_r - 2\omega^T\Xi_{\mathcal{F}}\bar{\phi} - 2\varpi_r^T\Xi\bar{\phi}\\ &\quad + c_1\omega^T\omega + c_2\varpi_r^T\varpi_r - \kappa_2\bar{\phi}^T\bar{\phi}\\ &\leq -\begin{bmatrix}\omega\\ \varpi_r\\ \bar{\phi}\end{bmatrix}^T\begin{bmatrix}\Pi_0 & -\Gamma_{\mathcal{F}} & \Xi_{\mathcal{F}}\\ -\Gamma_{\mathcal{F}}^T & \Pi & \Xi\\ \Xi_{\mathcal{F}}^T & \Xi^T & \kappa_2 I_{Nn}\end{bmatrix}\begin{bmatrix}\omega\\ \varpi_r\\ \bar{\phi}\end{bmatrix}\end{aligned} \tag{24}$$

According to (12), there exists constant $\epsilon_0 > 0$ such that

$$\begin{aligned}\dot{\mathcal{V}}_o &\leq -\epsilon_0\left\|\begin{bmatrix}\omega\\ \varpi_r\\ \bar{\phi}\end{bmatrix}\right\|^2\\ &\leq -\epsilon_0\left\|\begin{bmatrix}\omega\\ \varpi_r\end{bmatrix}\right\|^2\end{aligned} \tag{25}$$

Since $\varpi = \varpi_r + \varpi_c$ and $\varpi_c = 1_N \otimes \omega$, it holds that $\|\varpi_i\| \leq \|[\omega^T, \varpi_r^T]^T\|$, which indicates that

$$\|\varpi\| \leq \sqrt{N}\left\|\begin{bmatrix}\omega\\ \varpi_r\end{bmatrix}\right\| \tag{26}$$

Therefore, from (25)-(27), it implies that

$$\begin{aligned}\dot{\mathcal{V}}_o &\leq -\frac{\epsilon_0}{N}\|\varpi\|^2\\ &\leq -2\epsilon_1\mathcal{V}_o\end{aligned} \tag{27}$$

where $\epsilon_1 = \frac{\epsilon_0}{2N\max_{i\in\mathcal{N}}\{\lambda_{max}(\mathcal{F}_i)\}}$. Thus, $\mathcal{V}_o$ converges to zero with the decay late at least $2\epsilon_1$, it indicates that ϖ_i exponentially converges to zero. The proof is finished.

Remark 4. It is worth noting that the effectiveness of the distributed observer design relies on the observer gain matrices $\mathcal{Y}_i$, $\mathcal{F}_{io}$ and $\mathcal{F}_{iu}$ in LMI (12). To guarantee success of our proposed algorithm, the following Lemma 3 is introduced based on observable decomposition, which indicates the LMI (12) of Lemma 2 is feasible.

Lemma 3. *For the observable pair $(\mathcal{A}, col(\mathcal{J}_i)_{i\in\mathcal{N}})$, there exist constant $\sigma_0 > 0$ and matrices $\mathcal{F}_{io} > 0$, $\mathcal{F}_{iu} > 0$ and $\mathcal{H}_i$ satisfying the LMI (12).*

Proof. Define the matrix $\mathcal{Q}$ as

$$\mathcal{Q} = \begin{bmatrix}\mathcal{Q}_{11} & \mathcal{Q}_{12} & \mathcal{Q}_{13}\\ \mathcal{Q}_{12}^T & \mathcal{Q}_{22} & \mathcal{Q}_{23}\\ \mathcal{Q}_{13}^T & \mathcal{Q}_{23}^T & \mathcal{Q}_{33}\end{bmatrix} > 0 \tag{28}$$

where

$$\begin{cases}\mathcal{Q}_{11} = -\sum_{i=1}^{N}\Gamma_i - c_1 I_n, \quad \mathcal{Q}_{12} = -\Gamma_{\mathcal{F}},\\ \mathcal{Q}_{13} = \Xi_{\mathcal{F}}, \quad \mathcal{Q}_{22} = \sigma_0\sigma_1 I_{Nn} - \Gamma - c_2 I_{Nn},\\ \mathcal{Q}_{23} = \Xi, \quad \mathcal{Q}_{33} = \kappa_2 I_{Nn}\end{cases} \tag{29}$$

By using Schur complement [44], (28) is equivalent to

$$\mathcal{Q}_{33} > 0 \tag{30a}$$

$$\mathcal{Q}_{22} - \mathcal{Q}_{23}\mathcal{Q}_{33}^{-1}\mathcal{Q}_{23}^T > 0 \tag{30b}$$

$$\mathcal{Q}_{11} - \begin{bmatrix}\mathcal{Q}_{12} & \mathcal{Q}_{13}\end{bmatrix}\begin{bmatrix}\mathcal{Q}_{22} & \mathcal{Q}_{23}\\ \mathcal{Q}_{23}^T & \mathcal{Q}_{33}\end{bmatrix}^{-1}\begin{bmatrix}\mathcal{Q}_{12}^T\\ \mathcal{Q}_{13}^T\end{bmatrix} > 0 \tag{30c}$$

where $\mathcal{Q}_{33} = \kappa_2 I_{Nn}$, so (30a) holds. Because $\mathcal{Q}_{23}\mathcal{Q}_{33}^{-1}\mathcal{Q}_{23}^T$ does not rely on σ_0 and $\mathcal{Q}_{22} = \sigma_0\sigma_1 I_{Nn} - \Gamma - c_2 I_{Nn}$, σ_0 is sufficiently large such that (30b) is satisfied. In addition, for some $\bar{\epsilon}_1 > 0$,

$$\begin{bmatrix} \mathcal{Q}_{12} & \mathcal{Q}_{13} \end{bmatrix} \begin{bmatrix} \mathcal{Q}_{22} & \mathcal{Q}_{23} \\ \mathcal{Q}_{23}^T & \mathcal{Q}_{33} \end{bmatrix}^{-1} \begin{bmatrix} \mathcal{Q}_{12}^T \\ \mathcal{Q}_{13}^T \end{bmatrix} \leq \bar{\epsilon}_1 I_n \tag{31}$$

Now, to show that (30c) has solution, there exists $\mathcal{F}_{io} > 0$, $\mathcal{F}_{iu} > 0$ and $\mathcal{H}_i$ such that $\mathcal{Q}_{11} > \bar{\epsilon}_1 I_n$. Furthermore, recalling the definition of $\mathcal{Q}_{11}$ in (29), it indicates that

$$-\sum_{i=1}^{N} \Gamma_i > \bar{\epsilon}_2 I_n \tag{32}$$

where $\bar{\epsilon}_2 = \bar{\epsilon}_1 + c_1$.

Recalling the definition of Γ_i in (12), and the observability decomposition (7), it holds that

$$\Gamma_i = \mathcal{M}_i \begin{bmatrix} -\mathfrak{I}_{io} & \mathcal{A}_{ir}^T\mathcal{F}_{iu} \\ \mathcal{F}_{iu}\mathcal{A}_{ir} & \mathcal{F}_{iu}\mathcal{A}_{iu} + \mathcal{A}_{iu}^T\mathcal{F}_{iu} \end{bmatrix} \mathcal{M}_i^T \tag{33}$$

where $\mathfrak{I}_{io} = -\left(\mathcal{A}_{io}^T\mathcal{F}_{io} + \mathcal{F}_{io}\mathcal{A}_{io} + \mathcal{J}_{io}^T\left(\mathcal{H}_i + \mathcal{H}_i^T\right)\mathcal{J}_{io}\right)$.

Since $\mathcal{M}_i = [\mathcal{M}_{io}, \mathcal{M}_{iu}]$ given in (7), (33) can be rewritten as

$$\begin{aligned} &\mathcal{M}_i \begin{bmatrix} -\mathfrak{I}_{io} & \mathcal{A}_{ir}^T\mathcal{F}_{iu} \\ \mathcal{F}_{iu}\mathcal{A}_{ir} & \mathcal{F}_{iu}\mathcal{A}_{iu} + \mathcal{A}_{iu}^T\mathcal{F}_{iu} \end{bmatrix} \mathcal{M}_i^T \\ &\quad = -\mathcal{M}_{io}\mathfrak{I}_{io}\mathcal{M}_{io}^T + \mathcal{M}_{iu}\mathcal{F}_{iu}\mathcal{A}_{ir}\mathcal{M}_{io}^T \\ &\qquad + \mathcal{M}_{io}\mathcal{A}_{ir}^T\mathcal{F}_{iu}\mathcal{M}_{iu}^T \\ &\qquad + \mathcal{M}_{iu}\left(\mathcal{F}_{iu}\mathcal{A}_{iu} + \mathcal{A}_{iu}^T\mathcal{F}_{iu}\right)\mathcal{M}_{iu}^T \end{aligned} \tag{34}$$

Then, it can be concluded from (33)-(34) that

$$\begin{aligned} \sum_{i=1}^{N} \Gamma_i = &-\mathcal{M}_o\mathfrak{I}\mathcal{M}_o^T + \sum_{i=1}^{N}(\mathcal{M}_{iu}\mathcal{F}_{iu}\mathcal{A}_{ir}T_{io}^T + \mathcal{M}_{io}\mathcal{A}_{ir}^T\mathcal{F}_{iu}\mathcal{M}_{iu}^T) \\ &+ \sum_{i=1}^{N} \mathcal{M}_{iu}\left(\mathcal{F}_{iu}\mathcal{A}_{iu} + \mathcal{A}_{iu}^T\mathcal{F}_{iu}\right)\mathcal{M}_{iu}^T \end{aligned} \tag{35}$$

where $\mathcal{M}_o = \text{row}(\mathcal{M}_{io})_{i\in\mathcal{N}}$ and $\mathfrak{I} = \text{diag}(\mathfrak{I}_{io})_{i\in\mathcal{N}}$.

Before proceeding, the following Lemma 4 is needed.

Lemma 4. *If Assumption 3 is satisfied, it can be concluded that rank*$(\mathcal{M}_o) = n$*, namely,* $\mathcal{M}_o$ *has full row rank.*

Proof. See the Appendix A.

Based on Lemma 4, there exists a matrix $\tilde{\mathcal{M}}_o$ satisfying $\mathcal{M}_o\tilde{\mathcal{M}}_o = I_n$. It results from (35) that

$$\sum_{i=1}^{N} \Gamma_i = -\mathcal{M}_o(\mathfrak{I} - \Upsilon)\mathcal{M}_o^T \tag{36}$$

where $\Upsilon = \tilde{\mathcal{M}}_o\left(\sum_{i=1}^N\left(\mathcal{M}_{iu}\mathcal{F}_{iu}\mathcal{A}_{ir}\mathcal{M}_{io}^T + \mathcal{M}_{io}\mathcal{A}_{ir}^T\mathcal{F}_{iu}\mathcal{M}_{iu}^T\right) + \sum_{i=1}^N \mathcal{M}_{iu}\left(\mathcal{F}_{iu}\mathcal{A}_{iu} + \mathcal{A}_{iu}^T\mathcal{F}_{iu}\right)\mathcal{M}_{iu}^T\right)\tilde{\mathcal{M}}_o^T$.

Hence, according to (36), (32) can be rewritten as

$$\mathcal{M}_o(\mathfrak{I} - \Upsilon)\mathcal{M}_o^T > \bar{\epsilon}_2 I_n \tag{37}$$

Therefore, it is clear that $\mathcal{F}_{io}$ is positive definite such that the eigenvalues of $\mathfrak{I}_{io}$ and $\mathfrak{I}$ can be large enough. Thus, (37) is satisfied. In light of the above analysis, there exist $\mathcal{F}_{io}$, $\mathcal{F}_{iu}$ and $\mathcal{H}_i$ such that the LMI (28) is satisfied, which indicates the LMI (12) also holds. The proof is finished.

Remark 5. In (10), based on the observability decomposition given in (7), the state of the ith distributed observer is decomposed as the observable part of $(\mathcal{A}, \mathcal{J}_i)$ and the unobservable part. On the one hand, the observable part is stabilized by constructing appropriate gain matrices $\mathcal{Y}_i$ and $\mathcal{F}_{io}$. On the other hand, the unobservable part arrives a consensus with the estimates of neighbors' observers. For more information about this, we refer the reader to the previous work [12]. It should be emphasized that the feasibility of the LMI condition (12) about the designed distributed observer with OSL dynamics are one of the contributions of this article.

Note that all output or state information of the leader should be obtained by its neighbors (followers) in the design of the distributed observer with the linear dynamics of the leader [8,9]. However, neighbors of the leader can only measure partial output information $\mathcal{J}_i x_0$ in this article, which is milder condition than these works [8,9]. In addition, the leader with OSL nonlinear dynamics is considered, which is more general.

3.2. Design of distributed fault-tolerant control

In this subsection, to realize tracking control for the MASs affected by actuator faults, the distributed observer-based FT controller with performance guarantees is constructed as

$$\begin{cases} u_i = \bar{u}_i \hat{u}_i \\ \bar{u}_i = (I_n - \chi(\hat{\delta}_i))^{-1} \\ \hat{u}_i = \bar{\gamma}\mathcal{K}\eta_i - \frac{\psi_d(\|\eta_i\|_{\mathcal{D}})\hat{\vartheta}_i^2 \mathcal{B}^T\mathcal{D}\eta_i}{\psi_d(\|\eta_i\|_{\mathcal{D}})\hat{\vartheta}_i\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|+\mu_i} \end{cases} \tag{38}$$

with

$$\begin{cases} \dot{\hat{\delta}}_i = \beta_{i1}\,\mathrm{Proj}\left(\hat{\delta}_i, -\bar{\mu}_{i1}\hat{\delta}_i - \psi_d(\|\eta_i\|_{\mathcal{D}})\chi\left(u_i\right)\mathcal{B}_i^{\mathrm{T}}\mathcal{D}\eta_i\right) \\ \dot{\hat{\vartheta}}_i = -\beta_{i2}\bar{\mu}_{i2}\hat{\vartheta}_i + \beta_{i2}\psi_d(\|\eta_i\|_{\mathcal{D}})\left\|\eta_i^{\mathrm{T}}\mathcal{D}\mathcal{B}\right\| \end{cases} \tag{39}$$

where $\hat{\delta}_i$ and $\hat{\vartheta}_i$ are the estimation of δ_i and ϑ_i, respectively. $\bar{\gamma}$, β_{i1} and β_{i2} are positive design parameters. Continuous functions $\mu_i(t) > 0$, $\bar{\mu}_{i1}(t) > 0$ and $\bar{\mu}_{i2}(t) > 0$ satisfy $\lim_{t\to\infty}\int_0^t \mu_i(\varsigma)d\varsigma \le \tilde{\mu}$, $\lim_{t\to\infty}\int_0^t \bar{\mu}_{i1}(\varsigma)d\varsigma \le \tilde{\mu}$ and $\lim_{t\to\infty}\int_0^t \bar{\mu}_{i2}(\varsigma)d\varsigma \le \tilde{\mu}$, where $\tilde{\mu}$ is bounded positive constant. Proj$(\cdot)$ is the projection operator as given in [45]. $\mathcal{K} = -\mathcal{B}^T\mathcal{D}$ is gain matrix, where matrix $\mathcal{D}$ will be defined later. The tracking error η_i is defined as $\eta_i = x_i - \zeta_i$.

Remark 6. Compared with the FT control work for single OSL system in [46], the OSL MASs are considered in this article and our method has two significant features. Firstly, the term $\bar{u}_i$ in (38) is introduced to effectively reduce the number of adaptive laws, which can reduce the on-line computational burden and the difficulty of theory analysis. Secondly, the GRPF is introduced to improve the transient performance of OSL MASs affected by actuator faults.

Combining (4), (10) and (38), it can be concluded that

$$\begin{aligned} \dot{\eta}_i &= \dot{x}_i - \dot{\zeta}_i \\ &= \mathcal{A}x_i + \mathcal{B}(\Phi_i u_i + \rho_i) - \mathcal{A}\zeta_i - \mathcal{Y}_i\left(\mathcal{J}_i\zeta_i - \tilde{y}_i\right) \\ &\quad - \sigma_0 s_i \mathcal{F}_i^{-1}\sum_{j=1}^N a_{ij}\left(\zeta_j - \zeta_i\right) + \tilde{\phi}_i \\ &= \mathcal{A}x_i + \mathcal{B}(\chi(\tilde{\delta}_i)u_i + \hat{u}_i + \rho_i) - \mathcal{A}\zeta_i - \mathcal{Y}_i\left(\mathcal{J}_i\zeta_i - \tilde{y}_i\right) \\ &\quad - \sigma_0 s_i \mathcal{F}_i^{-1}\sum_{j=1}^N a_{ij}\left(\zeta_j - \zeta_i\right) + \tilde{\phi}_i \\ &= \mathcal{A}x_i + \bar{\gamma}\mathcal{B}\mathcal{K}\eta_i - \mathcal{A}\zeta_i + \mathcal{B}_i\rho_i - \mathcal{Y}_i\left(\mathcal{J}_i\zeta_i - \tilde{y}_i\right) \\ &\quad - \sigma_0 s_i \mathcal{F}_i^{-1}\sum_{j=1}^N a_{ij}\left(\zeta_j - \zeta_i\right) + \tilde{\phi}_i \\ &\quad + \mathcal{B}\chi(\tilde{\delta}_i)u_i - \frac{\psi_d(\|\eta_i\|_{\mathcal{D}})\hat{\vartheta}_i^2\mathcal{B}\mathcal{B}^T\mathcal{D}\eta_i}{\psi_d(\|\eta_i\|_{\mathcal{D}})\hat{\vartheta}_i\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\| + \mu_i} \\ &= (\mathcal{A} + \bar{\gamma}\mathcal{B}\mathcal{K})\eta_i + \mathcal{B}\rho_i + \mathcal{B}\chi(\tilde{\delta}_i)u_i \\ &\quad - \frac{\psi_d(\|\eta_i\|_{\mathcal{D}})\hat{\vartheta}_i^2\mathcal{B}\mathcal{B}^T\mathcal{D}\eta_i}{\psi_d(\|\eta_i\|_{\mathcal{D}})\hat{\vartheta}_i\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\| + \mu_i} - \mathcal{Y}_i\left(\mathcal{J}_i\zeta_i - \tilde{y}_i\right) \\ &\quad - \sigma_0 s_i \mathcal{F}_i^{-1}\sum_{j=1}^N a_{ij}\left(\zeta_j - \zeta_i\right) + \tilde{\phi}_i \end{aligned} \tag{40}$$

where $\tilde{\phi}_i = \phi(x_i) - \phi(\zeta_i)$.

Theorem 1. *If there exist matrix $\mathcal{D} = \mathcal{D}^T(\mathcal{D} > 0)$, positive constants $\bar{\kappa}_1$, $\bar{\kappa}_2$, τ_1 and $\bar{\gamma}$ satisfying*

$$\begin{bmatrix} \mathcal{Z}_1 & \mathcal{Z}_2 \\ \mathcal{Z}_2^T & -\bar{\kappa}_2 I_n \end{bmatrix} < 0 \tag{41}$$

where $\mathcal{Z}_1 = \mathcal{A}\hat{\mathcal{D}} + \hat{\mathcal{D}}\mathcal{A}^T - 2\bar{\gamma}\mathcal{B}\mathcal{B}^T + \bar{c}_1\hat{\mathcal{D}}^2 + \bar{c}_3\hat{\mathcal{D}}$, $\mathcal{Z}_2 = I_n + \bar{c}_2\hat{\mathcal{D}}$, $\bar{c}_1 = (\bar{\kappa}_1\gamma_1 + \bar{\kappa}_2\gamma_2 + \tau_1)$, $\bar{c}_2 = 0.5(-\bar{\kappa}_1 + \bar{\kappa}_2\gamma_3)$, $\hat{\mathcal{D}} = \mathcal{D}^{-1}$. (41) should be considered in the following three cases.

1. *If $\bar{c}_1 > 0$, then*
$$\begin{bmatrix} \mathcal{Z}_3 & \mathcal{Z}_2 & \mathcal{Z}_4 \\ \mathcal{Z}_2^T & -\bar{\kappa}_2 I_n & 0 \\ \mathcal{Z}_4^T & 0 & -I_n \end{bmatrix} < 0 \tag{42}$$
where $\mathcal{Z}_3 = \mathcal{A}\hat{\mathcal{D}} + \hat{\mathcal{D}}\mathcal{A}^T - 2\bar{\gamma}\mathcal{B}\mathcal{B}^T + \bar{c}_3\hat{\mathcal{D}}$, $\mathcal{Z}_4 = I_n + \bar{c}_2\hat{\mathcal{D}}$, $\mathcal{Z}_4 = \sqrt{\bar{c}_1}\hat{\mathcal{D}}$.
2. *If $\bar{c}_1 = 0$, then*
$$\begin{bmatrix} \mathcal{Z}_3 & \mathcal{Z}_2 \\ \mathcal{Z}_2^T & -\bar{\kappa}_2 I_n \end{bmatrix} < 0 \tag{43}$$
3. *If $\bar{c}_1 < 0$, then*
$\min \mathit{Trace}\left(\tilde{\mathcal{P}}\mathcal{P} + 0.5\tilde{\mathcal{Q}}\hat{\mathcal{Q}} - 0.5\bar{c}_1\tilde{\mathcal{Q}}\mathcal{P}^2\right)$
subject to
$$\left\{\begin{array}{l} \begin{bmatrix} \mathcal{Z}_3 - \hat{\mathcal{Q}} & \mathcal{Z}_2 \\ \mathcal{Z}_2^T & -\bar{\kappa}_2 I \end{bmatrix} < 0, \quad \begin{bmatrix} \mathcal{P} & I_n \\ I_n & \tilde{\mathcal{P}} \end{bmatrix} \geq 0 \\ \begin{bmatrix} \hat{\mathcal{Q}} & I_n \\ I_n & \tilde{\mathcal{Q}} \end{bmatrix} \geq 0, \quad \begin{bmatrix} -\bar{c}_1 I_n & \tilde{\mathcal{P}} \\ \tilde{\mathcal{P}}^T & \tilde{\mathcal{Q}} \end{bmatrix} \geq 0 \end{array}\right. \tag{44}$$
where $\hat{\mathcal{Q}} = -\bar{c}_1\mathcal{P}^2, \mathcal{P} = \tilde{\mathcal{P}}^{-1}, \hat{\mathcal{Q}} = \tilde{\mathcal{Q}}^{-1}$.

Then, the tracking error η_i satisfies the guaranteed performance bounds $\|\eta_i\|_{\mathcal{D}} < \theta(t)$. Moreover, the control objective can be arrived by employing the distributed observer-based FT controller (38) with adaptive law (39) and the distributed observer (10).

Proof. Define the Lyapunov function $\mathcal{V}_c^i$ as

$$\mathcal{V}_c^i = \psi\left(\|\eta_i\|_{\mathcal{D}}\right) + \frac{1}{\beta_{i1}}\tilde{\delta}_i^T\tilde{\delta}_i + \frac{1}{\beta_{i2}}\tilde{\vartheta}_i^2 \tag{45}$$

where $\mathcal{I}_\theta = \left\{\|\eta_i\|_{\mathcal{D}} : \|\eta_i\|_{\mathcal{D}} < \theta\right\}$, $\tilde{\delta}_i = \hat{\delta}_i - \delta_i$, $\tilde{\vartheta}_i = \hat{\vartheta}_i - \vartheta_i$, and

$$\begin{aligned} \frac{\mathrm{d}\psi\left(\|\eta_i\|_{\mathcal{D}}\right)}{\mathrm{d}t} &= \frac{\mathrm{d}\psi\left(\|\eta_i\|_{\mathcal{D}}\right)}{\mathrm{d}\|\eta_i\|_{\mathcal{D}}^2}\frac{\mathrm{d}\|\eta_i\|_{\mathcal{D}}^2}{\mathrm{d}t} + \frac{\mathrm{d}\psi\left(\|\eta_i\|_{\mathcal{D}}\right)}{\mathrm{d}\theta}\frac{\mathrm{d}\theta}{\mathrm{d}t} \\ &= 2\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\dot{\eta}_i - 2\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\|\eta_i\|_{\mathcal{D}}^2\frac{\dot{\theta}}{\theta} \end{aligned} \tag{46}$$

Then, it indicates from (40), (45) and (46) that

$$\begin{aligned} \dot{\mathcal{V}}_c^i &= \frac{\mathrm{d}\psi\left(\|\eta_i\|_{\mathcal{D}}\right)}{dt} + \frac{2}{\beta_{i1}}\tilde{\delta}_i^T\dot{\tilde{\delta}}_i + \frac{2}{\beta_{i2}}\tilde{\vartheta}_i\dot{\tilde{\vartheta}}_i \\ &= 2\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\dot{\eta}_i - 2\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\|\eta_i\|_{\mathcal{D}}^2\frac{\dot{\theta}}{\theta} + \frac{2}{\beta_{i1}}\tilde{\delta}_i^T\dot{\tilde{\delta}}_i + \frac{2}{\beta_{i2}}\tilde{\vartheta}_i\dot{\tilde{\vartheta}}_i \\ &= \psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\eta_i^T\left(2\mathcal{D}(\mathcal{A} + \bar{\gamma}\mathcal{B}\mathcal{K})\right)\eta_i - \frac{2\psi_d^2\left(\|\eta_i\|_{\mathcal{D}}\right)\hat{\vartheta}_i^2\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|^2}{\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\hat{\vartheta}_i\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\| + \mu_i} \\ &\quad + 2\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|\rho_i + 2\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\mathcal{B}\chi(\tilde{\delta}_i)u_i \\ &\quad + 2\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}(-\mathcal{Y}_i\left(\mathcal{J}_i\zeta_i - \tilde{y}_i\right) - \sigma_0 s_i\mathcal{F}_i^{-1}\sum_{j=1}^N a_{ij}\left(\zeta_j - \zeta_i\right)) \\ &\quad + 2\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\tilde{\phi}_i - 2\psi_d\left(\|\eta_i\|_{\mathcal{D}}\right)\|\eta_i\|_{\mathcal{D}}^2\frac{\dot{\theta}}{\theta} + \frac{2}{\beta_{i1}}\tilde{\delta}_i^T\dot{\tilde{\delta}}_i + \frac{2}{\beta_{i2}}\tilde{\vartheta}_i\dot{\tilde{\vartheta}}_i \end{aligned} \tag{47}$$

It results from (47) that

$$
\begin{aligned}
-2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\Theta_i &\leq \psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\tau_1\eta_i^T\eta_i+\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\tau_2\|\Theta_i\|^2\\
&\leq \psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\tau_1\eta_i^T\eta_i+\bar{\epsilon}_6 e^{-\bar{\epsilon}_3 t}
\end{aligned}
\tag{48}
$$

with $\Theta_i=\mathcal{Y}_i\left(\mathcal{J}_i\zeta_i-\tilde{y}_i\right)+\sigma_0 s_i\mathcal{F}_i^{-1}\sum_{j=1}^{N}a_{ij}\left(\zeta_j-\zeta_i\right)$, $\tau_2=\frac{\|\mathcal{D}\|^2}{\tau_1}$, according to Lemma 2, $\|\Theta_i\|^2\leq\bar{\epsilon}_4 e^{-\bar{\epsilon}_3 t}$, $\bar{\epsilon}_6=\bar{\epsilon}_4\bar{\epsilon}_5\tau_2$. $\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)$ is bounded and $\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\leq\bar{\epsilon}_5$, for $\mathcal{I}_\theta=\left\{\|\eta_i\|_{\mathcal{D}}:\|\eta_i\|_{\mathcal{D}}<\theta\right\}$. $\bar{\epsilon}_3$, $\bar{\epsilon}_4$ and $\bar{\epsilon}_5$ are bounded positive constants.

Substituting (39) and (48) into (47), it results that

$$
\begin{aligned}
\dot{\mathcal{U}}_c^i \leq\;& \psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\eta_i^T(\mathcal{D}\mathcal{A}+\mathcal{A}^T\mathcal{D}-2\bar{\gamma}\mathcal{D}\mathcal{B}\mathcal{B}^T\mathcal{D}+\tau_1 I+\bar{c}_3\mathcal{D})\eta_i\\
&-\frac{2\psi_d^2\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\hat{\vartheta}_i^2\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|^2}{\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\hat{\vartheta}_i\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|+\mu_i}+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|\vartheta_i\\
&+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\mathcal{B}\chi(\tilde{\delta}_i)u_i+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\tilde{\phi}_i\\
&+2\tilde{\delta}_i^T\left(\mathrm{Proj}(\hat{\delta}_i,-\bar{\mu}_{i1}\hat{\delta}_i-\psi_d(\|\eta_i\|_{\mathcal{D}})\chi\left(u_i\right)\mathcal{B}_i^T\mathcal{D}\eta_i)\right)\\
&-2\bar{\mu}_{i2}\tilde{\vartheta}_i\hat{\vartheta}_i+2\psi_d(\|\eta_i\|_{\mathcal{D}})\tilde{\vartheta}_i\left\|\eta_i^{\mathrm{T}}\mathcal{D}\mathcal{B}\right\|+\bar{\epsilon}_6 e^{-\bar{\epsilon}_3 t}
\end{aligned}
\tag{49}
$$

where $\bar{c}_3=2\max_{t\geq 0}\{\|\frac{\dot{\theta}}{\theta}\|\}$.

Moreover, it can be concluded from (49) that

$$
\begin{aligned}
&-\frac{2\psi_d^2\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\hat{\vartheta}_i^2\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|^2}{\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\hat{\vartheta}_i\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|+\mu_i}+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|\rho_i+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|\tilde{\vartheta}_i\\
&\leq-\frac{2\psi_d^2\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\hat{\vartheta}_i^2\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|^2}{\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\hat{\vartheta}_i\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|+\mu_i}+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|\vartheta_i+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|\tilde{\vartheta}_i\\
&\leq-\frac{2\psi_d^2\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\hat{\vartheta}_i^2\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|^2}{\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\hat{\vartheta}_i\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|+\mu_i}+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\left\|\eta_i^T\mathcal{D}\mathcal{B}\right\|\hat{\vartheta}_i\\
&\leq 2\mu_i
\end{aligned}
\tag{50}
$$

where $\|\rho_i\|\leq\vartheta_i$.

In light of the definition of projection operator [45] and (49), it indicates that

$$
\begin{aligned}
&2\tilde{\delta}_i^T\,\mathrm{Proj}\left(\hat{\delta}_i,-\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\chi(u_i)\mathcal{B}^T\mathcal{D}\eta_i-\bar{\mu}_{i1}\hat{\delta}_i\right)\\
&+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\mathcal{B}\chi(\tilde{\delta}_i)u_i\\
&\quad=2\tilde{\delta}_i^T\,\mathrm{Proj}\left(\hat{\delta}_i,-\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\chi(u_i)\mathcal{B}^T\mathcal{D}\eta_i-\bar{\mu}_{i1}\hat{\delta}_i\right)\\
&\qquad+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\mathcal{B}\chi(u_i)\tilde{\delta}_i\\
&\quad=2\tilde{\delta}_i^T\,\mathrm{Proj}\left(\hat{\delta}_i,-\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\chi(u_i)\mathcal{B}^T\mathcal{D}\eta_i-\bar{\mu}_{i1}\hat{\delta}_i\right)\\
&\qquad+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\tilde{\delta}_i^T\chi\left(u_i\right)\mathcal{B}^T\mathcal{D}\eta_i\\
&\quad=2\tilde{\delta}_i^T(\mathrm{Proj}\left(\hat{\delta}_i,-\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\chi\left(u_i\right)\mathcal{B}^T\mathcal{D}\eta_i-\bar{\mu}_{i1}\hat{\delta}_i\right)\\
&\qquad+\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\chi\left(u_i\right)\mathcal{B}^T\mathcal{D}\eta_i+\bar{\mu}_{i1}\hat{\delta}_i)-2\bar{\mu}_{i1}\tilde{\delta}_i^T\hat{\delta}_i\\
&\quad\leq-2\bar{\mu}_{i1}\tilde{\delta}_i^T\hat{\delta}_i
\end{aligned}
\tag{51}
$$

Note that

$$
\begin{aligned}
-2\bar{\mu}_{i1}\tilde{\delta}_i^T\hat{\delta}_i-2\bar{\mu}_{i2}\tilde{\vartheta}_i\hat{\vartheta}_i &=-2\bar{\mu}_{i1}\tilde{\delta}_i^T(\tilde{\delta}_i+\delta_i)-2\bar{\mu}_{i2}\tilde{\vartheta}_i(\tilde{\vartheta}_i+\vartheta_i)\\
&\leq\frac{\bar{\mu}_{i1}\|\tilde{\delta}_i\|^2}{2}+\frac{\bar{\mu}_{i2}\tilde{\vartheta}_i^2}{2}
\end{aligned}
\tag{52}
$$

Substituting (50)-(52) into (49) satisfies

$$
\begin{aligned}
\dot{\mathcal{U}}_c^i \leq\;& \psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\eta_i^T(\mathcal{D}\mathcal{A}+\mathcal{A}^T\mathcal{D}-2\bar{\gamma}\mathcal{D}\mathcal{B}\mathcal{B}^T\mathcal{D}+\tau_1 I+\bar{c}_3\mathcal{D})\eta_i\\
&+2\psi_d\left(\left\|\eta_i\right\|_{\mathcal{D}}\right)\eta_i^T\mathcal{D}\tilde{\phi}_i+\upsilon
\end{aligned}
\tag{53}
$$

where $\upsilon=\frac{\bar{\mu}_{i1}\|\bar{\mu}_i\|^2}{2}+\frac{\bar{\mu}_{i2}\tilde{\vartheta}_i^2}{2}+\bar{\epsilon}_6 e^{-\bar{\epsilon}_3 t}+2\mu_i$.

According to Definitions 1 and 2 and Assumption 2, it is clear that

$$\bar{\kappa}_1\gamma_1\eta_i^T\eta_i - \bar{\kappa}_1\eta_i^T\tilde{\phi}_i \geq 0 \tag{54a}$$

$$-\bar{\kappa}_2\tilde{\phi}_i^T\tilde{\phi}_i + \bar{\kappa}_2\gamma_2\eta_i^T\eta_i + \bar{\kappa}_2\gamma_3\eta_i^T\tilde{\phi}_i \geq 0 \tag{54b}$$

Moreover, for $\mathcal{I}_\theta = \left\{\|\eta_i\|_\mathcal{D} : \|\eta_i\|_\mathcal{D} < \theta\right\}$, $\psi_d\left(\|\eta_i\|_\mathcal{D}\right) > 0$ together with (54a) and (54b), it holds that

$$\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\left(\bar{\kappa}_1\gamma_1\eta_i^T\eta_i - \bar{\kappa}_1\eta_i^T\tilde{\phi}_i\right) \geq 0 \tag{55a}$$

$$\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\left(-\bar{\kappa}_2\tilde{\phi}_i^T\tilde{\phi}_i + \bar{\kappa}_2\gamma_2\eta_i^T\eta_i + \bar{\kappa}_2\gamma_3\eta_i^T\tilde{\phi}_i\right) \geq 0 \tag{55b}$$

Based on (53), (55a) and (55b), it can be concluded that

$$\begin{aligned}
\dot{\mathcal{U}}_c^i \leq{}& \psi_d\left(\|\eta_i\|_\mathcal{D}\right)\eta_i^T\left(\mathcal{D}\mathcal{A} + \mathcal{A}^T\mathcal{D} - 2\bar{\gamma}\mathcal{D}\mathcal{B}\mathcal{B}^T\mathcal{D} + \tau_1 I + \bar{c}_3\mathcal{D}\right)\eta_i \\
&+ \bar{\kappa}_1\gamma_1\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\eta_i^T\eta_i - \bar{\kappa}_1\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\eta_i^T\tilde{\phi}_i - \bar{\kappa}_2\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\tilde{\phi}_i^T\tilde{\phi}_i \\
&+ \bar{\kappa}_2\gamma_2\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\eta_i^T\eta_i + \bar{\kappa}_2\gamma_3\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\eta_i^T\tilde{\phi}_i \\
&+ 2\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\eta_i^T\mathcal{D}\tilde{\phi}_i + \upsilon \\
\leq{}& \psi_d\left(\|\eta_i\|_\mathcal{D}\right)\eta_i^T\left(\mathcal{D}\mathcal{A} + \mathcal{A}^T\mathcal{D} - 2\bar{\gamma}\mathcal{D}\mathcal{B}\mathcal{B}^T\mathcal{D} + \tau_1 I + \bar{c}_3\mathcal{D}\right)\eta_i \\
&+ \bar{\kappa}_1\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\begin{bmatrix}\eta_i \\ \tilde{\phi}\end{bmatrix}^T\begin{bmatrix}\gamma_1 I_n & \left(-\frac{I_n}{2}\right) \\ \left(-\frac{I_n}{2}\right) & 0\end{bmatrix}\begin{bmatrix}\eta_i \\ \tilde{\phi}_i\end{bmatrix} \\
&+ \bar{\kappa}_2\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\begin{bmatrix}\eta_i \\ \tilde{\phi}_i\end{bmatrix}^T\begin{bmatrix}\gamma_2 I_n & \frac{\gamma_3 I_n}{2} \\ \frac{\gamma_3 I_n}{2} & (-I_n)\end{bmatrix}\begin{bmatrix}\eta_i \\ \tilde{\phi}_i\end{bmatrix} \\
&+ 2\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\eta_i^T\mathcal{D}\tilde{\phi}_i + \upsilon \\
\leq{}& \psi_d\left(\|\eta_i\|_\mathcal{D}\right)\begin{bmatrix}\eta_i \\ \tilde{\phi}_i\end{bmatrix}^T\Lambda\begin{bmatrix}\eta_i \\ \tilde{\phi}_i\end{bmatrix} + \upsilon
\end{aligned} \tag{56}$$

where $\Lambda = [\Lambda_{11}, \Lambda_{12}; \Lambda_{12}^T, \Lambda_{22}]$ with $\Lambda_{11} = \mathcal{D}\mathcal{A} + \mathcal{A}^T\mathcal{D} - 2\bar{\gamma}\mathcal{D}\mathcal{B}\mathcal{B}^T\mathcal{D} + \bar{c}_1 I_n + \bar{c}_3\mathcal{D}$, $\Lambda_{12} = \mathcal{D} + \bar{c}_2 I_n$, $\Lambda_{22} = -\bar{\kappa}_2 I_n$. Let $\bar{\mathcal{D}} = \mathrm{diag}(\hat{\mathcal{D}}, I_n)$. If $\Lambda < 0$, it also holds that

$$\bar{\mathcal{D}}\Lambda\bar{\mathcal{D}} = \begin{bmatrix}\mathcal{Z}_1 & \mathcal{Z}_2 \\ \mathcal{Z}_2^T & -\bar{\kappa}_2 I_n\end{bmatrix} < 0 \tag{57}$$

According to (41), (56) and (57), there exists a constant $\tau_3 > 0$ such that

$$\begin{aligned}
\dot{\mathcal{U}}_c^i \leq{}& -\tau_3\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\left\|\begin{bmatrix}\eta_i \\ \tilde{\phi}_i\end{bmatrix}\right\|^2 + \upsilon \\
\leq{}& -\tau_3\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\|\eta_i\|^2 + \upsilon \\
\leq{}& -\frac{\tau_3}{\tau_4}\left(\psi_d\left(\|\eta_i\|_\mathcal{D}\right)\psi\left(\|\eta_i\|_\mathcal{D}\right) - \frac{1}{2}\psi\left(\|\eta_i\|_\mathcal{D}\right)\right) \\
&- \frac{\tau_3}{2\tau_4}\psi\left(\|\eta_i\|_\mathcal{D}\right) + \upsilon \\
\leq{}& -\frac{\tau_3}{2\tau_4}\psi\left(\|\eta_i\|_\mathcal{D}\right) + \upsilon
\end{aligned} \tag{58}$$

where $\tau_4 = \max\{\lambda(\mathcal{D})\}$.

Then, it can be obtained from (58) that

$$\mathcal{U}_c^i(t) \leq \mathcal{U}_c^i(0) - \frac{\tau_3}{2\lambda_2}\int_0^t \psi\left(\|\eta_i\|_\mathcal{D}\right)d\varsigma + \int_0^t \upsilon d\varsigma \tag{59}$$

It indicates from (59) that $\lim_{t\to\infty}\frac{\tau_3}{2\tau_4}\int_0^t \frac{\|\eta_i\|_\mathcal{D}^2}{\theta^2(\varsigma) - \|\eta_i\|_\mathcal{D}^2}d\varsigma \leq \mathcal{U}_c(0) + \bar{\upsilon}$. In addition, $\lim_{t\to\infty}\int_0^t \upsilon d\varsigma < \bar{\upsilon}$, $\bar{\upsilon} > 0$ is a bounded constant. Thus, $\lim_{t\to\infty}\|\eta_i\|_\mathcal{D}^2 = 0$ and $\|\eta_i\|_\mathcal{D} < \theta(t)$. Since $\|\eta_i\|_\mathcal{D}^2 \geq \tau_4\|\eta_i\|^2$, it indicates that $\lim_{t\to\infty}\eta_i = 0$. According to Lemma 2, it indicates that $\lim_{t\to\infty}(\zeta_i - x_0) = 0$, then $\lim_{t\to\infty}(x_i - x_0) = \lim_{t\to\infty}(x_i - \zeta_i) + \lim_{t\to\infty}(\zeta_i - x_0) = 0$. The proof is completed.

Remark 7. Note that the distributed observer is introduced to not only reconstruct the state of the leader but also avoid the phenomenon of faults propagation among followers. Followers broadcast estimated state of the distributed observer rather than the real state of followers to its neighbors over communication network, so the faults of the real agents will not have effect on the dynamics

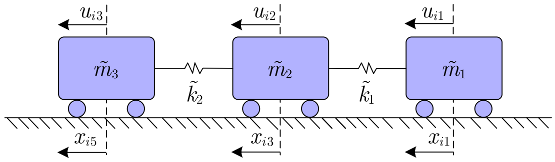

Fig. 1. Three-mass-spring system.

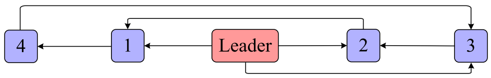

Fig. 2. Communication graph.

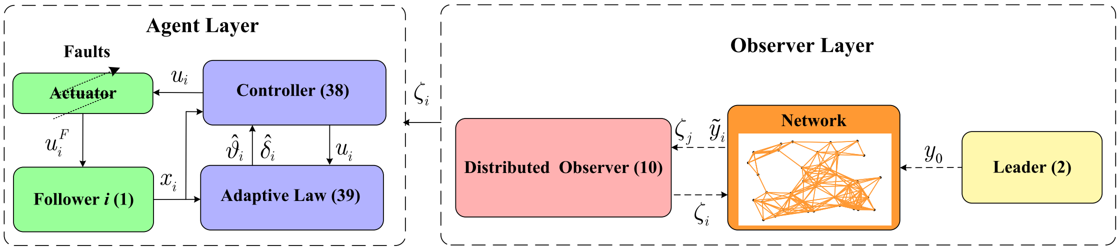

Fig. 3. Schematic of the proposed method.

of neighbor agents. It should be mentioned that the actuator faults might lead to a sudden change for MASs, to improve the transient performance of the MASs, the GRPF $\psi\left(\|\eta_i\|_{\mathcal{D}}\right)$ is introduced to guarantee $\|\eta_i\|_{\mathcal{D}}$ within the bounds θ. Different from FT control works [37,39], neighbors of the leader can only have part output information rather than all state information in this article. In addition, the OSL function existing in the dynamics of the leader will increase the difficulty of the distributed observer design, which is a hard topic.

4. Simulation results

4.1. Three-mass-spring system

In this section, a three-mass-spring system arisen from [47] is considered to illustrate the proposed algorithm. Here, the three-mass-spring system is depicted in Fig. 1, and its dynamics are described by

$$\dot{x}_i = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{\tilde{k}_1}{\tilde{m}_1} & 0 & \frac{\tilde{k}_1}{\tilde{m}_1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{\tilde{k}_1}{\tilde{m}_2} & 0 & -\frac{\tilde{k}_1+\tilde{k}_2}{\tilde{m}_2} & 0 & \frac{\tilde{k}_2}{\tilde{m}_2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{\tilde{k}_2}{\tilde{m}_3} & 0 & -\frac{\tilde{k}_2}{\tilde{m}_3} & 0 \end{bmatrix} x_i + \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{\tilde{m}_1} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{\tilde{m}_2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{\tilde{m}_3} \end{bmatrix} \begin{bmatrix} u_{i1} \\ u_{i2} \\ u_{i3} \end{bmatrix} + \phi(x_i) \tag{60}$$

where $x_i = [x_{i1}, x_{i2}, x_{i3}, x_{i4}, x_{i5}, x_{i6}]^T$, x_{i1}, x_{i3}, x_{i5} and x_{i2}, x_{i4}, x_{i6} represent displacements and velocities separately, $\tilde{k}_1$ and $\tilde{k}_2$ are spring constants, $\tilde{m}_h$ denotes mass, u_{ih} is the control input, for $h = 1, 2, 3$. Two springs connect three mass cars as three-mass-spring system, which moves in the horizontal direction. The leader's control input is zero. All followers are required to track the trajectory of the leader. Based on the proposed controller (38), all followers can realize the tracking task. The communication graph is given in Fig. 2, which satisfies Assumption 1. The proposed distributed observer-based fault-tolerant control scheme is graphically presented in Fig. 3. It can be seen from Fig. 3 that the proposed method is divided into two parts: 1) The observer layer is introduced to design the distributed observer (10) under JOC. 2) The purpose of the agent layer is to track the observer state ζ_i based on the controller (38). For illustration, $\bar{\mathcal{J}}$ and $\mathcal{J}_i$ are considered as

$$\bar{\mathcal{J}} = \begin{bmatrix} \mathcal{J}_1 \\ \hline \mathcal{J}_2 \\ \hline \mathcal{J}_3 \end{bmatrix} = \left[\begin{array}{cccccc} 1 & 0 & -1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 & -1 & 0 \end{array}\right] \tag{61a}$$

$$\mathcal{J}_4 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{61b}$$

According to (61a), (61b), it indicates that the pair $(\mathcal{A}, \mathcal{J}_i)$ is unobservable, while $(\mathcal{A}, \mathrm{col}(\mathcal{J}_i)_{i\in\mathcal{N}})$ is observable. Then, based on (61a), (61b) and the observability decomposition, choose the orthogonal matrix $\mathcal{M}_i = [\mathcal{M}_{io}, \mathcal{M}_{iu}]$ as

$$\mathcal{M}_{1o} = \begin{bmatrix} -\frac{2}{\sqrt{6}}I_2 & 0 \\ \frac{1}{\sqrt{6}}I_2 & -\frac{1}{\sqrt{2}}I_2 \\ \frac{1}{\sqrt{6}}I_2 & \frac{1}{\sqrt{2}}I_2 \end{bmatrix}, \quad \mathcal{M}_{1u} = \begin{bmatrix} \frac{1}{\sqrt{3}}I_2 \\ \frac{1}{\sqrt{3}}I_2 \\ \frac{1}{\sqrt{3}}I_2 \end{bmatrix}, \quad \mathcal{M}_{2o} = \begin{bmatrix} \frac{1}{\sqrt{2}}I_2 & 0 \\ 0 & -I_2 \\ \frac{1}{\sqrt{2}}I_2 & 0 \end{bmatrix}, \quad \mathcal{M}_{2u} = \begin{bmatrix} -\frac{1}{\sqrt{2}}I_2 \\ 0 \\ \frac{1}{\sqrt{2}}I_2 \end{bmatrix}$$

and $\mathcal{M}_{3o} = \mathcal{M}_{1o}$, $\mathcal{M}_{3u} = \mathcal{M}_{1u}$, $\mathcal{M}_4 = \mathcal{M}_{4u} = I_6$. Let parameters $\tilde{k}_1 = \tilde{k}_2 = \tilde{k}_3 = 1N/m$ and $\tilde{m}_1 = \tilde{m}_2 = \tilde{m}_3 = 1kg$. Besides, $\phi(x_i) = [0,0,0,0,0,-0.1\mathrm{sign}(x_{i6})\sqrt{|x_{i6}|}]^T$. In light of Definitions 1 and 2, it can be obtained that $\gamma_1 = -0.01$, $\gamma_2 = 0$, $\gamma_3 = -0.01$. Parameters in LMI conditions (12) and (41) are chosen as $\sigma_0 = 200$, $c_1 = -0.04$, $c_2 = -0.01$, $c_3 = -10.5$, $\bar{c}_1 = 0$, $\bar{c}_2 = -1.505$, $\bar{c}_3 = 0.1$, $s_i = 1$, for $i = 1,\ldots,4$. Parameters in controller (38) and adaptive law (39) are chosen as $\bar{\gamma} = 1$, $\theta = 2.4e^{-0.1t} + 0.05$, $\beta_{i1} = 0.1$, $\beta_{i2} = 1$, $\mu_i = 0.5e^{-0.15t}$, $s_i = 1$, for $i = 1,\ldots,4$. Since we do not have a hardware platform, all of the simulations are carried out on a Windows 11, Intel Core i7-13700k and 3.40 GHz computer with 64 GB RAM and tested in MATLAB 2022b. By using YALMIP toolbox and SDPT3 solver in MATLAB 2022b, the feasible solutions for LMI conditions (12) and (41) can be computed as

$$[\mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Y}_3, \mathcal{Y}_4] = \begin{bmatrix} -0.6457 & -8.6741 & -0.1352 & 0 \\ -0.1190 & 1.5316 & 0.0875 & 0 \\ 0.5105 & -10.3210 & -0.5105 & 0 \\ 0.2065 & -7.0954 & -0.2065 & 0 \\ 0.1352 & -8.6741 & 0.6457 & 0 \\ -0.0875 & 1.5316 & 0.1190 & 0 \end{bmatrix}$$

and $\mathcal{H}_1 = \mathcal{H}_3 = -7.1772$, $\mathcal{H}_2 = -16.2143$, $\mathcal{H}_4 = 0$. Moreover, $\mathcal{F}_i$, $\mathcal{D}$, and $\mathcal{K}$ are listed as follows:

$$\mathcal{F}_1 = \mathcal{F}_3 = \begin{bmatrix} 7.5478 & -1.1810 & -4.4884 & 0.0630 & -1.6630 & -0.6743 \\ -1.1810 & 10.7943 & 0.0630 & 2.0668 & -0.6743 & -1.7643 \\ -4.4884 & 0.0630 & 10.3732 & -1.9182 & -4.4884 & 0.0630 \\ 0.0630 & 2.0668 & -1.9182 & 6.9632 & 0.0630 & 2.0668 \\ -1.6630 & -0.6743 & -4.4884 & 0.0630 & 7.5478 & -1.1809 \\ -0.6743 & -1.7643 & 0.0630 & 2.0668 & -1.1809 & 10.7943 \end{bmatrix}$$

$$\mathcal{F}_2 = \begin{bmatrix} 8.2151 & -2.8012 & -4.3647 & -1.3058 & -2.2270 & 1.2519 \\ -2.8012 & 10.6373 & 1.3527 & 1.9413 & 1.2519 & -1.3034 \\ -4.3647 & 1.3527 & 10.8554 & -2.2495 & -4.3647 & 1.3527 \\ -1.3058 & 1.9413 & -2.2495 & 7.3029 & -1.3058 & 1.9413 \\ -2.2270 & 1.2519 & -4.3647 & -1.3058 & 8.2151 & -2.8012 \\ 1.2519 & -1.3034 & 1.3527 & 1.9413 & -2.8012 & 10.6373 \end{bmatrix}$$

$$\mathcal{F}_4 = \begin{bmatrix} 7.5994 & -1.0280 & -4.3647 & -1.3058 & -1.6113 & -0.5213 \\ -1.0280 & 10.9463 & 1.3527 & 1.9413 & -0.5213 & -1.6124 \\ -4.3647 & 1.3527 & 10.8554 & -2.2495 & -4.3647 & 1.3527 \\ -1.3058 & 1.9413 & -2.2495 & 7.3029 & -1.3058 & 1.9413 \\ -1.6113 & -0.5213 & -4.3647 & -1.3058 & 7.5994 & -1.0280 \\ -0.5213 & -1.6124 & 1.3527 & 1.9413 & -1.0280 & 10.9463 \end{bmatrix}$$

$$\mathcal{D} = \begin{bmatrix} 1.7761 & 0.5239 & -0.5008 & 0.0863 & 0.2849 & 0.1503 \\ 0.5239 & 1.2414 & 0.0863 & 0.1482 & 0.1503 & 0.0553 \\ -0.5008 & 0.0863 & 2.5619 & 0.5880 & -0.5008 & 0.0863 \\ 0.0863 & 0.1482 & 0.5880 & 1.1485 & 0.0863 & 0.1482 \\ 0.2849 & 0.1503 & -0.5008 & 0.0863 & 1.7761 & 0.5239 \\ 0.1503 & 0.0553 & 0.0863 & 0.1482 & 0.5239 & 1.2414 \end{bmatrix}$$

$$\mathcal{K} = \begin{bmatrix} -0.5239 & -1.2414 & -0.0863 & -0.1482 - 0.1503 - 0.0553 \\ -0.0863 & -0.1482 & -0.5880 & -1.1485 - 0.0863 - 0.1482 \\ -0.1503 & -0.0553 & -0.0863 & -0.1482 - 0.5239 - 1.2414 \end{bmatrix}$$

In simulation, the actuator faults occur at 20 s, i.e., $\delta_1 = \mathrm{diag}(0.6, 0.4, 0.3)$, $\delta_2 = \mathrm{diag}(0.2, 0.5, 0.6)$, $\delta_3 = \mathrm{diag}(0.3, 0.4, 0.5)$, $\delta_4 = \mathrm{diag}(0.6, 0.3, 0.4)$, $\rho_1 = [2.2 + 0.1\sin(t); -2.2 - 0.1\sin(t); 2.8 - 0.1\sin(t)]$, $\rho_2 = [-3.1 + 0.11\sin(t); -1.7 - 0.11\sin(t); 3 + 0.11\sin(t)]$, $\rho_3 = [1.6 + 0.11\sin(t); -1.9 + 0.11\sin(t); -3.3 + 0.11\sin(t)]$, $\rho_4 = [-2.6 - 0.11\sin(t); 2.9 - 0.1\sin(t); 2.8 - 0.11\sin(t)]$. Simulation results are given in Figs. 4–6. It can be seen from Fig. 4 and Fig. 5 that the distributed state observer converges to the state of the leader. The curves of $\|\eta_i\|_{\mathcal{D}}$ and θ are shown in Fig. 6, where noises are considered. It can be seen that $\|\eta_i\|_{\mathcal{D}}$ can be constrained within θ under actuator faults and noises. To further demonstrate the advantages of the designed control strategy, the curves of $\|\eta\|$ with the proposed method, the fault-tolerant method in [37,46] and the method in [29] are shown in Fig. 7. From Fig. 7, it is shown that after faults occur, compared with the curves of $\|\eta\|$ with the proposed method and the fault-tolerant method in [37,46], the curve

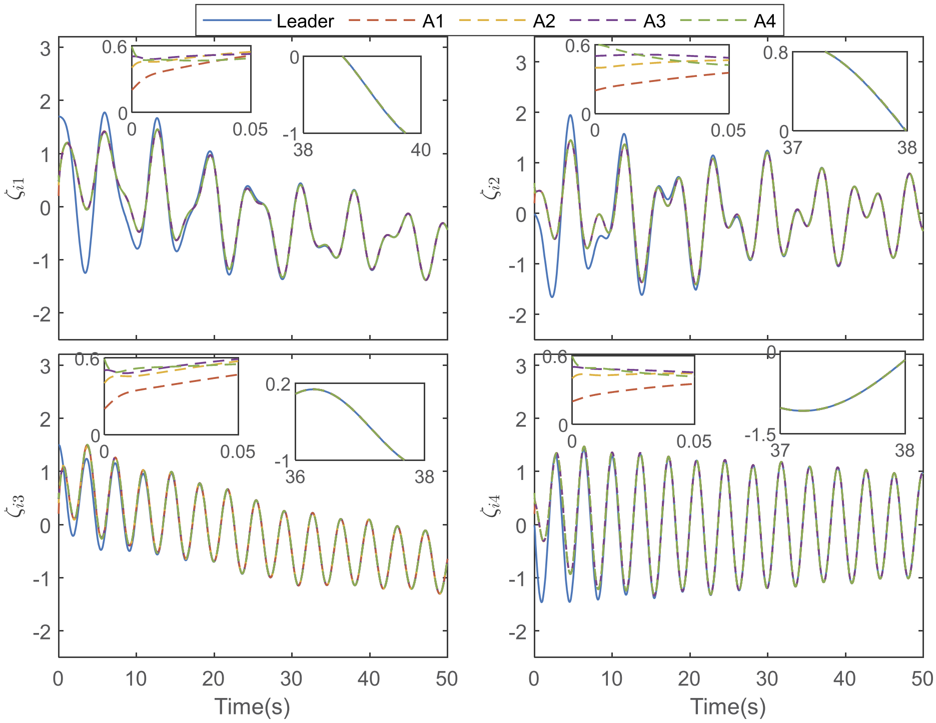

Fig. 4. Curves of ζ_{if}, $i = 1, 2, 3, 4$ and $f = 1, 2, 3, 4$. Ai denotes ith follower.

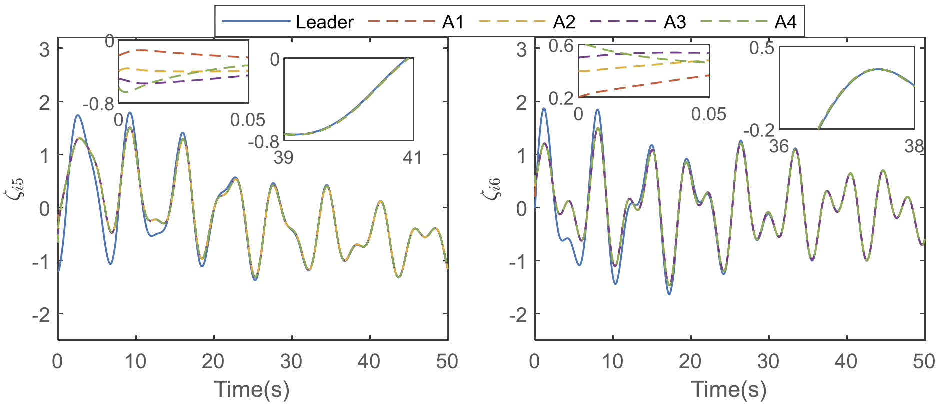

Fig. 5. Curves of ζ_{if}, $i = 1, 2, 3, 4$ and $f = 5, 6$. Ai denotes ith follower.

of $\|\eta\|$ with the method [29] exhibits the highest amplitude. In addition, the transient performance of MASs experiences a sharp deterioration following the occurrence of faults, primarily due to the omission of actuator faults considerations in [29]. Compared with [29], the fault-tolerant control scheme is introduced to deal with actuator faults, the transient performance of [37,46] is quite better than that of [29]. It should be noted that in [37,46], the effect of actuator faults is slowly compensated and the transient performance is quite poor, compared with the proposed method. Since the proposed method considers both actuator faults and guaranteed performance, the effect of actuator faults is quickly compensated. After faults occur, the proposed method has the best transient performance compared with the methods in [29,37,46].

4.2. Discussion

1) In the distributed observer (10), the feedback matrices $\mathcal{Y}_i$ and $\mathcal{F}_i$ are determined by LMI condition (12), $i = 1, 2, 3, 4$. The computational complexity of LMI condition (12) is $\mathcal{O}(N^2 n^2)$, which is dependent on both the dimension of the leader state n, and the number of agents N. The computational complexity of LMI condition (12) is same as the distributed observer results for linear systems under JOC in [12]. If the number of agents is very large, then the computational complexity is very high. To reduce complexity, a feasible method is to divide agents into two groups, namely group $\mathfrak{R}_1$ and group $\mathfrak{R}_2$. In group $\mathfrak{R}_1$, the communication graph is strong

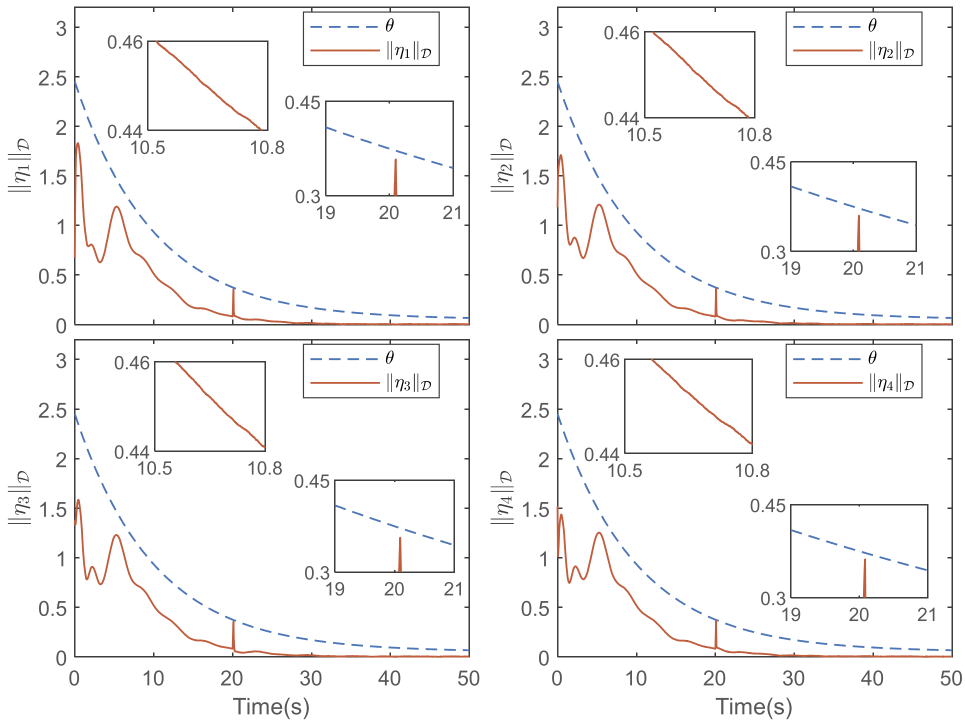

Fig. 6. Curves of $\|\eta_i\|_\mathcal{D}$ with noises, $i = 1, 2, 3, 4$.

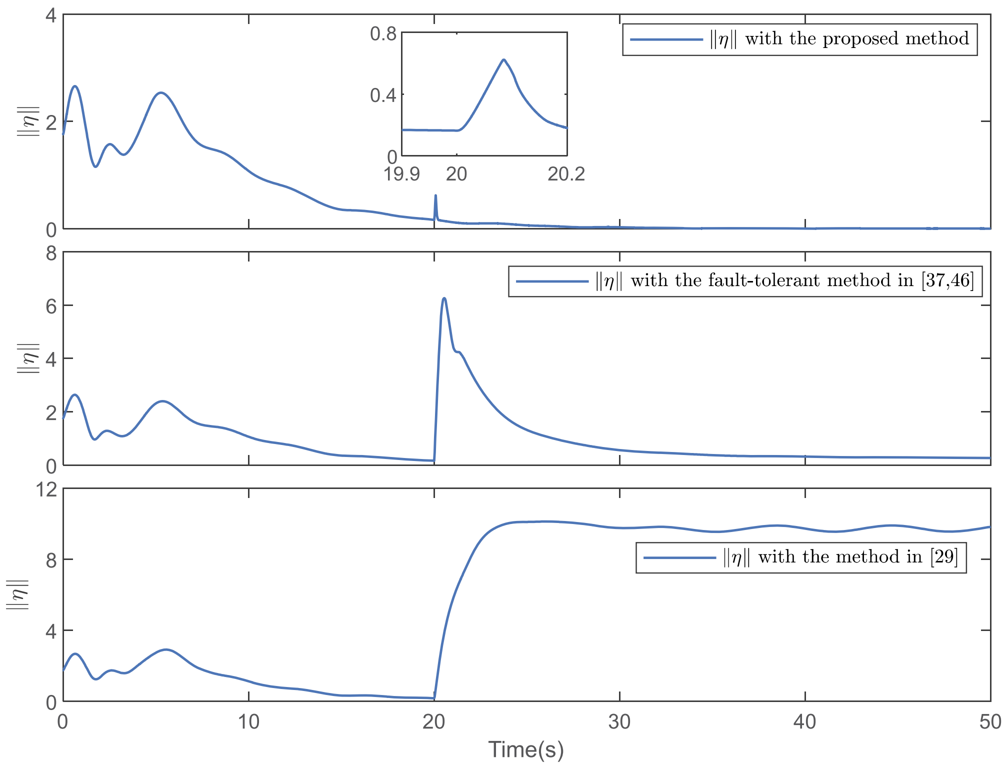

Fig. 7. Curves of $\|\eta\|$ under the proposed method, the fault-tolerant method in [37,46] and the method in [29], where $\eta = \mathrm{col}(\eta_i)_{i\in\mathcal{N}}$.

connected, each agent can obtain the partial output information of the leader under JOC. In group $\mathfrak{R}_2$, the communication graph is connected and there exists at least one path from group $\mathfrak{R}_1$ to group $\mathfrak{R}_2$. In general, the dimension of the system output is less than or equal to the dimension of the state. To satisfy JOC, there are at most n agents in group $\mathfrak{R}_1$. By doing this, the computational complexity of LMI condition (12) will be less than or equal to $\mathcal{O}(n^4)$. Besides, for group $\mathfrak{R}_2$, the computational complexity of LMI condition is $\mathcal{O}(n^2)$. Future research will focus on how to reduce computational complexity.

2) The proposed distributed observer (10) and fault-tolerant controller (38) are dependent on the feedback matrices $\mathcal{Y}_i$ and $\mathcal{K}$, $i = 1, 2, 3, 4$, which are determined by LMI conditions (12) and (41). It is worth noting that the LMI conditions (12) and (41) rely on the known matrices $\mathcal{A}$ and $\mathcal{B}$. If we consider that there exists parametric uncertainty in $\mathcal{A}$ or $\mathcal{B}$, the feedback matrices obtained by nominal $\mathcal{A}$ and $\mathcal{B}$ might not guarantee that the signals of error dynamics (11) and (40) are bounded. In this article, the three-mass-spring system is considered, due to prolonged operation or aging, the constants $\tilde{k}_1$ and $\tilde{k}_2$ of springs might undergo changes, which will cause that $\mathcal{A}$ is uncertain. How to extend the proposed method to MASs with parametric uncertainty (like uncertainty in $\mathcal{A}$) is a topic for future research.

5. Conclusion

In this article, the fault-tolerant performance guaranteed tracking control problem for OSL MASs under JOC has been investigated. Compared with the previous works, the studied problem is more general in the sense that only partial output information of the leader can be obtained by its neighbors, and the dynamics of agents contain OSL nonlinearities. Firstly, we have designed the distributed OSL observers via a decomposition technique for the error space and provided lemmas for the solvability of such observers via JOC. Then, by virtue of the proposed distributed observer, a distributed adaptive fault-tolerant protocol has been developed. It is shown that all followers asymptotically track the leader and the predefined performance can also be guaranteed, even in the presence of actuator faults. By using the proposed algorithm, actuator faults of each agent will not have effect on the dynamics of its neighbors over the communication graph. In further work, the problem of parameter uncertainty will be considered, which would make this method more applicable to realistic scenarios.

CRediT authorship contribution statement

Wang Yang: Term, Conceptualization, Methodology, Writing-Original Draft, Writing-Review & Editing, Software, Validation; **Jiuxiang Dong:** Term, Conceptualization, Methodology, Writing-Review & Editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Proof of the Lemma 4

Proof. For $\mathcal{M}_{io}$, it is clear that $\mathrm{im}(\mathcal{M}_{io}) = \ker(\mathcal{X}_i)^{\perp}$, where $\mathcal{X}_i = \mathrm{col}(\mathcal{J}_i\mathcal{A}^{\varrho-1})_{\varrho\in\bar{\mathcal{N}}}$, $\bar{\mathcal{N}} = \{1, \dots, n\}$. Then, it can be concluded that

$$\begin{aligned}
\mathrm{im}(\mathcal{M}_o)^{\perp} &= \left(\sum_{i=1}^{N}\mathrm{im}(\mathcal{M}_{io})\right)^{\perp} \\
&= \cap_{i=1}^{N}\mathrm{im}(\mathcal{M}_{io})^{\perp} \\
&= \cap_{i=1}^{N}\ker(\mathcal{X}_i) \\
&= \ker\left(\mathrm{col}(\mathcal{X}_i)_{i\in\mathcal{N}}\right) \\
&= 0
\end{aligned} \tag{62}$$

Hence, $\mathrm{rank}(\mathcal{M}_o) = n$, the proof is finished.

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